VERIFICATION REPORT FOR “IMPROVED COOKSTOVE GROUPED PROJECT IN PAPUA NEW GUINEA”

Report ID	4416 – VE
Project title	Improved Cookstove Grouped Project in Papua New Guinea
Project ID	4416
Verification period	23/09/2024 to 31/12/2024
Original date of issue	21/04/2025
Most recent date of issue	24/07/2025
Version	1.0
***VCS Standard* Version**	4.7[1]
Client	TEM PNG Cookstoves Project Pte. Ltd.
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Approved by	Indrapal Parmar
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Summary:

A description of the verification of the project

The 1st Verification and methodology changes (in line with VCS methodology change project description deviation[4.0]) of the project title "Improved Cookstove Grouped Project in Papua New Guinea' VCS ID – 4416, have been undertaken by SustainCERT with accordance to the relevant VCS standard, V4.7[1], VCS Program Guide, V4.4[2]. Relevant requirements of the ISO 14064-2 and ISO 14064-3, as well as applicable criteria for consistent project operations, monitoring and reporting have been applied for verification.

The project VCS 4416 involves dissemination of Improved cook stoves in Papua New Guinea, primarily targeting low-income households to promote cleaner and more efficient cooking solutions. The project is implemented by TEM PNG Cookstoves Project Pte. Ltd., which also serves as the project proponent. Carbon finance underpins the project's strategy, enabling local engagement in stove distribution, household registration, monitoring, and maintenance activities. The ICSs distributed, manufactured under the "Burn" brand, feature a stainless-steel insulated combustion chamber with thermal efficiency exceeding 25%[18]. These stoves are engineered to significantly reduce firewood usage and indoor air pollution by improving combustion efficiency and heat transfer compared to traditional three-stone fires.

The Purpose and scope of verification

Purpose:

The verification service provided by SustainCERT for the project activity VCS 4416 is to perform independent periodic verification for the monitoring period 23/09/2024 to 31/12/2024 and validate the revised PDD against the new methodology VM0050 V1.0[08]. The verification of the project is an independent review and ex-post determination by SustainCERT of the monitored reductions in GHG emissions that have occurred, against the VCS standard, V4.7[1], VCS Program Guide, V4.4[2], the Project baseline and host party criteria.

Scope of verification

The scope of verification is to establish/verify that:

- The project activity has been implemented and operated as per the approved registered and revised PD[5/8].

- All the physical features (project technology) of the project are in place in accordance with registered PD and MR[/5/].
- The revised PDD, monitoring report[/8/]and other supporting documents provided are complete in accordance with the latest applicable version of VCS standard and methodology change procedure.
- The actual monitoring systems and procedures comply with the monitoring systems and procedures described in the monitoring plan, any registered monitoring plan[/5/], the approved methodology[/7/] including applicable tool(s) and/or, where applicable, the approved standardized baseline.
- The data recorded and stored as per the monitoring methodology including applicable tool(s) and, where applicable, the standardized baseline

The Monitoring Period:

The project activity VCS 4416 is undergoing 1st verification from 23/09/2024 to 31/12/2024 (both days included), under its ten-year fixed crediting period from 23/09/2024 to 22/09/2034[/6/].

The Method and Criteria used for Verification.

Desk Review:

Review of project information, monitoring plan specifically monitoring frequency, quality assurance and quality control system against the registered PD[/5/] and applicable VCS standard and applied methodology VCS Methodology from Sectoral Scope 3 –VM0050 Energy Efficiency and Fuel-Switch Measures in Cookstoves, V1.0[/08/].

Onsite audit Interview with the relevant personnel to confirm the implementation of the project by paying particular attention to the operational and data collection procedures, in accordance with the registered PD[/5/] and methodology.

- A detailed cross check of the information provided in the monitoring report against the relevant evidence such as data sources and technical specification of measuring equipment.
- A detailed cross check of the calculations made for the estimation of greenhouse gas emissions.
- An evaluation of data management and the quality assurance and quality control system in the context of their influence on the generation and reporting of emissions reduction.

The number of findings raised during verification:

A risk-based approach has been followed for the verification of the project. During verification, 04 CARs, 16 CL and 00 FAR were raised during verification.

All the issues raised were resolved successfully, details of which can be found in appendix 4 of the report.

Any uncertainties associated with the verification:

The assessment team thoroughly reviewed the revised PDD[/09/] along with ex-ante calculation sheet, monitoring report[/8/] and the emission reduction calculation sheet[/9/] against the relevant evidence and applicable VCS requirements, VCS standard V4.7[/1/] and VCS guideline V4.4. It has been concluded that there are no more uncertainties associated with the project.

Summary of the verification conclusion:

The review of the revised PDD/09/ Monitoring Report/08/, emission reduction calculation sheet, supporting documentation, and the interview with relevant personnel during the on-site audit provided the assessment team with sufficient evidence to confirm compliance with the applicable criteria. SustainCERT confirms that the project activity, VCS 4416, has been successfully implemented in accordance with the registered Project Description and the applied methodology, VM0050: Energy Efficiency and Fuel-Switch Measures in Cookstoves, Version 1.0, in alignment with all relevant VCS Program requirements. SustainCERT confirms that the monitoring period from 23/09/2024 to 31/12/2024 (inclusive) has resulted in the generation of 13,443tCO_2e of emission reductions through the reduction in non-renewable biomass combustion.

The management of TEM PNG Cookstoves Project Pte. Ltd. (Project Proponent) is responsible for the preparation of the GHG emissions data and the reported GHG emission reductions, based on the procedures established in the project's Monitoring Plan outlined in the registered Project Design Document and the applied methodology.

CONTENTS

1 INTRODUCTION

1.1 Objective

The 1st Verification and methodology change/[26/] of the project titled "Improved Cookstove Grouped Project in Papua New Guinea" VCS ID – 4416, have been undertaken by SustainCERT, as requested by TEM PNG Cookstoves Project Pte Ltd project proponent for the monitoring period 23/09/2024 to 31/12/2024 (including both the days). The verification of project activity is the 1st periodic verification for the crediting period 23/09/2024 to 22/09/2034. The purpose of the verification is to conduct an independent review of the project information and confirm:

- The project activity is implemented in accordance with registered monitoring plan as mentioned in the initial Project Description[/5/] and revised PDD.
- All physical features (project technology) are in place.
- The actual monitoring systems and procedures comply with the monitoring systems and procedures described in the registered monitoring plan[/5/], the new applied methodology[/7/] including applicable tool(s) and/or, where applicable, the approved standardized baseline.
- The data collection procedures and records are as per the monitoring methodology VM0050 Energy Efficiency and Fuel-Switch Measures in Cookstoves, V1.0[/8/]

The verification followed the requirements mentioned in the VCS standard, version 4.7[/1/] and VCS guidelines, V4.4[/2/] ensuring the quality and consistency of the report.

1.2 Scope and Criteria

The scope of verification for this project activity involves assessing the claims made by the project proponent in the revised PDD along with ax- ante calculation sheet, monitoring report[/8/], emission reduction calculation sheet[/9/] and other supporting evidence (as mentioned in appendix 2) made available to the verifier for this monitoring period from 23/09/2024 to 31/12/2024, in accordance to following documents:

- VCS standard, V4.7[/1/]
- VCS program guideline, V4.4[/2/]
- VCS Methodology from Sectoral Scope 3 –VM0050 Energy Efficiency and Fuel-Switch Measures in Cookstoves, V1.0.[/7/]
- VCS Methodology change project description Deviation v4.0[/26/]

- Other relevant rules and requirements, including host part requirements/31/.

A risk-based approach has been followed for the verification of the project activity. Identifying and assessing the high-risk area and ensuring the reliability of the project's emission reductions generated.

The principles of accuracy, completeness, relevance, reliability and credibility were combined with a conservative approach to establish a traceable and transparent verification opinion. The verification considers both quantitative and qualitative information on emission reductions. The verification is not meant to provide any consultancy towards the client. However, stated requests for clarifications, corrective and/or forward actions may provide input for improvement of the monitoring activities.

1.3 Level of Assurance

The verification of the project activity is conducted by doing a detailed assessment of Monitoring report/8/, revised PDD against already registered PDD, emission reduction calculation sheet/9/ and all the relevant documents, and Distribution Database'/11/ (as mentioned in appendix 2) of the report against the applied methodology VM0050 Energy Efficiency and Fuel-Switch Measures in Cookstoves, V1.0./7/ .

The level of assurance achieved for the project verification falls under a reasonable level of assurance. A reasonable level of assurance is a high level of assurance regarding material misstatement, but not an absolute one.

Enough evidence was gathered by the assessment team to reduce the risk associated with the audit process. This means that there is some uncertainty arising from the use of sampling since it is possible that a material misstatement is missed.

1.4 Summary Description of the Project

This is the first monitoring report for the "Improved Cookstove Grouped Project in Papua New Guinea" (VCS ID 4416), implemented by TEM PNG Cookstoves Project Pte. Ltd. as a Grouped Project aimed at disseminating fuel-efficient improved cookstoves (ICS) to local communities across Papua New Guinea. Along with the verification, there is also a methodology change taking place in the project, from VMR0006, 1.2 to VM0050, V1.0, through the VERRA procedure VCS Methodology change project description Deviation, V4.0/26/

The ICS disseminated under this GP are high-efficiency Burn cookstoves, designed with stainless-steel insulated combustion chambers that improve fuel combustion and heat transfer efficiency/18//12/. These stoves reduce greenhouse gas emissions by lowering the quantity of firewood required for cooking, thereby minimizing non-renewable biomass consumption.

During the current monitoring period, a total of 19,206 ICS units were distributed as individual project instances across households in the Southern Highlands Province. Since the start of the project on 23/09/2024, these 19,206 stoves mark the first phase of implementation as verified through the interview with PP. Technical specifications of the ICS technology were cross-verified through the Project Design Document, manufacturer documentation, and onsite inspection during field visits to end-user households.

An onsite audit for the project activity was conducted from 17/02/2025 to 20/02/2025. The start date of the project activity is 23/09/2024, as confirmed through the interview with the Project Proponent, verification of the distribution database, and carbon wavier for the first ICS distributed[/21/]. The crediting period for the project activity is from 23/09/2024 to 22/09/2034, as verified from the revised Project Description Document, inline with VERRA rules and requirements.

Based on this, the assessment team confirms that the project activity, VCS 4416: Improved Cookstove Grouped Project in Papua New Guinea, has been implemented and is operational in accordance with the applied methodology VM0050: Energy Efficiency and Fuel-Switch Measures in Cookstoves, Version 1.0 (Applicability criteria related to methodology can be found in the sections below), the registered PD, the VCS Standard v4.7, and applicable VCS guidelines. The emission reduction calculations of 13,443tCO2e for the monitoring period from 23/09/2024 to 31/12/2024 have been found to be conservative and appropriate upon verification.

2 VERIFICATION PROCESS

The group Project activity has been undergoing 1st Verification under its fixed crediting period and methodology changed as VCS procedure for methodology change through project description deviation[/26/], the approach adapted to ensure the quality and credibility of emission reduction is described in the following sections.

2.1 Method and Criteria

The method and criteria used for verification consist of following phases.

A risk-based approach has been applied for the process.

Desk review:

- A document review of the revised PDD, monitoring report[/8/], emission reduction calculation sheet[/9/] and other evidence like monitoring survey data[/12/], signed user agreements[/14/], project distribution database[/11/] (as mentioned in Appendix 2 of the report) were strategically reviewed against the applied methodology VM0050, V1.0[/8/] and risk assessment was undertaken.

- Project activity sources contributing to the leakage as well as project emissions are assessed inline with applied methodology[/08/] .

- The frequency of measurements, QA/QC procedures and other relevant documents are verified.

  An onsite audit for the project was conducted from 17/02/2025 to 20/02/2025. Further details regarding the process have been mentioned in section 2.3 and 2.4 of the report.

- Interviews of the project representatives were conducted to discuss the implementation and operational status of the project with respect to the registered monitoring plan[/5/].

- Discussion of QA/QC procedure, data collection, storage and transfer.

- Interview and visual observation on the condition of the project technology stove, baseline stoves, perceived opinion on SDG claims of the projects, grievances if any

- Monitoring period claimed (from 23/09/2024 to 31/12/2024) was verified from the ER calculation[/9/], as well as from revised PDD, registered PDD and all the similar to projects on different registries to ensure, there is no double counting[/24/].

  Resolution of all the issues and findings raised, and the final verification statement was issued.

2.2 Document Review

As part of verification of the project, the primary activity performed was the strategic review and risk assessment of the documents submitted using a dedicated protocol. A detailed review of revised PDD, MR[/8/] and ER calculation[/9/] was performed to check:

- The completeness of the information with reference to the register PD[/5/] and registered PD.

- Review of project information, monitoring plan specifically monitoring frequency, quality assurance and quality control system against the registered PD[/5/] and applicable VCS standard and applied methodology version 1.0.

- A review of the QA/QC procedures was conducted to make sure the implementation is as per the register PD[/5/].

- Project specific assessment against applicable national framework and legislation requirement[/31/32/33/].

All the other evidence reviewed is mentioned in Appendix 2 of this report. The cross checks between information provided in the Monitoring report[/8/], VCS PD[/5/] and information from sources other than those used, if available, the team's sectoral or local expertise and, if necessary, independent background investigations.

2.3 Interviews

During this verification, an on-site visit was performed by the assessment team.

Site visit was emphasized on project technology user with visit to each household for visual observation as well as verbal confirmation on parameter of:

- Project technology usage by kitchen observation and interview with end users
- Project technology model and overall condition
- Stove repair mechanism (refer CL 12)
- Project unique identification barcode
- Baseline stove and baseline fuel
- Confirmation on signing end user agreements
- Perception on smoke level of project technology vs baseline stove

Project technology users interviewed on site for household visit can be found in the table below.

S.No	End user name and serial number	Type of correspondent	Ward	Details discussed	Date of onsite Visit – Auditor
1	Helen Peter FDYRH3	Household (batch 1)	Beechwood	Operationality of Project stove, Kitchen observation to understand if project ICS stove was recently used, usage rate, Benefits of using project stoves and related SDGs impact, Repair/maintenance aspect, Feeback opinion on project and related	18/02/2025 – Mr. Hayden Wagia
2	Relindah James FDYXZA	Household (batch 1)	Koropangi		
3	Kindeh yakala FDYRPB	Household (batch 1)	Kumuge		
4	Michael kapu ADYXF9	Household (batch 1)	koropang		
5	Penny tony NDYH9Y	Household (batch 1)	Kendal 4		

6	Nema Apa DYYRQ	Household (batch 1)	Station	communication channels to address the grievances, baseline stove usage, baseline scenario	
7	Aobu Mark ADYUER	Household (batch 2)	Yebi 2		19/02/2024 – Mr. Hayden Wagia
8	Patric Yama ADYUNE	Household (batch 2)	Yebi 1		
9	Tende kelely ADYSR7	Household (batch 2)	Yombi 2		
10	Issac kepo FDYXPC	Household (batch 2)	Yombi 1		
11	Luke Maka ADYGPE	Household (batch 2)	Nagop		

Based on the interviews conducted with project beneficiaries during the onsite visit and kitchen observations by the verifier team, it can be concluded that project ICS are actively used by the households. (CL 15 and CL 16 were raised related to the onsite visit can be found in Appendix 4 of the report). The project activity also involves the change in the methodology from VMR0006, V1.2 to VM0050, V1.0 During the onsite visit the baseline for the project activity was also checked to verify the continued applicability of the existing baseline as per the registered PDD.

Below tables outline the personnel involved in the interviews, along with their respective roles. The interviews specifically targeted individuals responsible for monitoring the project activity, data collection and management, as well as those involved in the quality assurance and quality control (QA/QC) procedures. The tables serve to identify the individuals interviewed and provide relevant information regarding their roles within the project.

Name	Designation	Topic Discussed	Name of the auditor
Geogina Tembi	Implementation partner		Mr. Hayden Wagia
Benlee Jessie	Implementation partner		

Eric Ulu	Implementation partner	On ground project implementation, Beneficiary identification, Project stove distribution, grievance mechanism, stakeholder consultation, employee wellbeing, training and employment	
Miclael Kisombo	In country manager	Project implementation, QA and QC procedure, Training, employment, Project stove distribution, grievance mechanism , stakeholder identification, repair mechanism, monitoring survey and baseline survey, ICS usage awareness	Muskan Chawla Mr. Hayden Wagia
Brian James	Implementation Partner		Mr. Hayden Wagia
Samuel Wamo	Stakeholder and baseline stove users	Stakeholder consultation, baseline survey, Baseline stove usage, baseline fuel usage, baseline fuel collection, grievance mechanism	Mr. Hayden Wagia
Maria iki	Stakeholder and baseline stove users		
Jenenifer Andy	Stakeholder and baseline stove users		
Gabriel Landea	Stakeholder and baseline stove users		

Tiks Rema	Stakeholder and baseline stove users		
Philip Kera	Stakeholder and baseline stove users	Stakeholder consultation, grievance mechanism, engagement process and awareness, employment, feedback mechanism, baseline scenario and stove usage in baseline	
Ruth Alalo	Stakeholder and baseline stove users		

Interview with the above personnel verified that baseline established during the initial validation of the project remains valid during the methodology change and is found to be inline with the new applied methodology VM0050, V1.0 section 6.

The key personnel interviewed during the opening meeting and closing meeting session of the onsite audit, and the main topics of the interviews are summarized in the tables below:

Table 2: Project proponent and Project implementation field team interviewed from 17/02/2025 to 20/02/2025.

S.No	**Name of the person Interviewed**	**Affiliation**	**Details Discussed**	**Audit Team**
1	Dr. Gabriel Machovsky Capuska	TEM	Project monitoring survey execution, project distribution mechanism, data collection (distribution database), grievances mechanism	Muskan Chawla Mr. Hayden Wagia
2	Cherie Sim	TEM	Project monitoring survey execution, project distribution mechanism, data collection (distribution database), grievances mechanism	

3	Shameer Muhammed	TEM	Project monitoring survey execution, project distribution mechanism, data collection (distribution database), grievances mechanism	

During the validation of the project a FAR was raised, “In Line with Section 4.1.12 of the VCS Standard v4.5, during the first verification of the project activity, the next verifying VVB shall conduct an on-site visit of the project activity to reassess the following information provided by the project proponent: Project Location, Baseline Scenario, Stakeholder Engagement and consultation.” Based on the FAR raised, the assessment team confirms that a comprehensive reassessment was carried out during the first verification. The following aspects were specifically addressed:

1. Project location: The VVB conducted an on-site audit and confirmed that all visited stoves were located within the designated project boundary in Papua New Guinea.

Verification was further supported by:

- KML files submitted by PP, detailing the geographical boundary.
- The distribution database, which included household specific coordinates and stove installation data.

2. Baseline Scenario: Assessment team directly verified the baseline scenario by visiting baseline households. Additionally, During the project household interviews, end users confirmed that their baseline technology consisted of inefficient open fire stoves. These findings were corroborated using a structured baseline questionnaire designed to gather data on pre-project cooking practices and technologies.

3.Stakeholder engagement and consultation: The VVB conducted interviews with:

The project proponent and its representatives,

Baseline users and stakeholders,

Implementation partners (local employees also part of stakeholders)

end users of the project cookstove and stakeholders

Interviews were conducted with all the above people and details related to stakeholders engagement and continuous grievance mechanism were discussed. Details related to feedback on stakeholders were also discussed during the onsite to verify the details related to stakeholder engagement.

Additionally, to the above information desk review was also conducted based on the evidence submitted by the PP for stakeholder engagement process, including records of community consultations, training sessions, and awareness activities.

Based on the he desk review, field verification, and interviews conducted, the assessment teams confirms that the FAR raised during validation has been fully addressed, and that the project location, baseline scenario, and stakeholder engagement have been reassessed and confirmed in accordance with the applicable standard required.

On the basis of the information collected through above interviews and then further cross checked with the evidence submitted in the Appendix 2 of the report, helped assessment team reach to the conclusion the project is inline with the applied methodology[/09/] and VCS rules and requirement, thus, a reasonable level of assurance is met.

2.4 Site Visits

The VVB carried out an on-site visit from 17/02/2025 to 20/02/2025 and physically inspected the project technical design and implementation as specified in the registered PD/5/, revised PDD and applied methodology VM0050, V1.0.

The site visit was conducted to verify the accuracy and completeness of the project implementation. The views were obtained during site observation and interactions with project proponent were considered while concluding the verification opinion.

Duration of the Onsite-Audit: 17/02/2025 to 20/02/2025			
S.No	**Assessment Criteria**	**Means of Verification**	**Assessment Opinion**
1	The assessment of the implementation and operation of the registered project activity	Registered PD/5/, revised PD with applied methodology VM0050, V1.0 Interviews with PP and onsite visit/10/	The information mentioned with reference to project implementation, efficiency and commissioning in MR is found consistent with the documents and information received during interviews.
2	A review of the monitoring survey for generating, aggregating and reporting the monitoring parameters.	Monitoring survey/12/ and emission reduction/9/ calculation sheet. Interviews with households/10/	The information related to the monitoring survey generating, aggregating and reporting the monitoring parameters mentioned in MR is found to be consistent with documents crosschecked and information received by interviews.
3	Interviews with relevant personnel to determine whether the operational and data collection procedures are implemented in accordance with the monitoring plan in the PD/5/	Interviews with PP/10/ Training records/22/	All the information related to operational and data collection is found to be inline with the PD/5/ and MR/8/.
4	A check of the monitoring equipment including data transfer and observations of	Interviews with PP/10/	Monitoring practices were verified with interview from PP, and it has been

	monitoring practices against the requirements of the PD, the applied methodology including the tools applied		observed that all the practices are found to be in line with the PD.
6	A review on calculation of emission reductions	Monitoring survey/12/ and emission reduction/9/	The values mentioned for ER calculation are found to be in line with the monitoring survey/12/. Methods, formulae, and emission factors for calculating baseline emissions have been followed in accordance with the applied methodology/8/.
7	An identification of quality control and quality assurance procedures in place to prevent or identify and correct any errors or omissions in the reported monitoring parameters	Interviews with PP/10/	Methods, formulae, and emission factors for calculating baseline emissions have been followed in accordance with the applied methodology/7/. All the QA/AC procedures were also verified on-site.

The sampling plan adopted by the VVB for carrying out the verification assessment is as follows.
It was also assessed that some areas of PNG is also considered high risk areas due to security reasons (https://www.smartraveller.gov.au/destinations/pacific/papua-new-guinea), Sample selection was also done considering the risk analysis in the area, however, it was made sure samples are covered from the entire population covering both the batches.

Project stove:
The Sampling and surveys for CDM project activities and programmes of activities (Version 9.0)[/18/] states under paragraph 28 that “When the project participants or the coordinating/managing entity have applied a sampling approach, the VVB may apply acceptance sampling as described in the steps indicated in paragraphs 29–38 below as part of validation/verification activities”.

Since the PP has monitored parameters (Operational status of the stove and usage rate of the stove) through surveys, Assessment team conducted acceptance sampling in line with paragraph 30 and 31 of the sampling standard version 9.0[/18/].

The Assessment team selected random sample of PP's monitoring survey records to check the acceptability (or otherwise) of the data for each such record with PP's sample records and determined if the PP's sample records meet the requirements.

Sample Size:

AQL	UQL	Producer Risk	Consumer Risk	Sample Size; Min	Acceptance No.
0.5%	20%	10%	10%	11	0

The Assessment team covered a total of 11 samples across from ICS with each ICS itself a single project instances to confirm monitoring survey results.

All the households interviewed during the site visit were found to have inefficient baseline stove stoves using firewood in the baseline and confirmed that the project stove was operational. Fair emission reduction calculations are performed based on appropriately recorded usage during the monitoring survey[/12/] for the current monitoring period. The Assessment team would like to conclude that based on acceptance sampling approach the project participant's sample record meets the requirement.

2.5 Resolution of Findings

As discussed above, a risk-based approach has been applied for the verification of the project activity. During the verification, the assessment team identified the issues which could impact the accuracy and the credibility of the emission reduction claimed by the project. The detailed analysis of the issues raised the approach used to resolve these issues can be found in Appendix 4 of this report.

The Corrective Action Request (CAR) was raised if:

- Modifications to the implementation, operation, and monitoring of the registered project activity have not been sufficiently documented by the project participants.
- Mistakes have been made in applying assumptions, data, or calculations of emission reductions that will impact the number of emission reductions.

The Clarification Request (CL) was raised if:

- Information is insufficient or not clear enough to determine whether the applicable requirements have been met.

The Forward Action Request (FAR) was raised if:

- If the monitoring and reporting require attention and/or adjustment for the next verification period.

During verification a total of 04 CARs, 16 CLs and 00 FAR were raised. The details of all the issues communicated to the PD can be found in Appendix 4. The issues raised were closed successfully.

2.5.1 Forward Action Requests

No forward action request has been raised for this monitoring period. However, there has a FAR from previous verification:
FAR - In Line with Section 4.1.12 of the VCS Standard v4.5, during the first verification of the project activity, the next verifying VVB shall conduct an on-site visit of the project activity to reassess the following information provided by the project proponent: Project Location, Baseline Scenario, Stakeholder Engagement and consultation.
Response - Based on the Raised FAR a site visit was conducted by the verification team more details related to which can be found in section 2.2 and 2.3 of this report.

2.6 Eligibility for Validation Activities

During the current monitoring period, the project underwent a methodology update, transitioning from VMR0006: Methodology for Installation of High Efficiency Firewood Cookstoves v1.1 to VM0050: Energy Efficiency and Fuel-Switch Measures in Cookstoves v1.0, through a Project Description deviation.
As required under VCS rules, a validation of the revised Project Description against the updated methodology was necessary. Accordingly, the VVB confirms that it conducted a validation activity in conjunction with the verification, as documented in Section 3.2 of the Verification Report.
The VVB, SustainCERT S.A., holds active accreditation under the VCS Program for both validation and verification scopes, as confirmed by the Verra VVB eligibility listing[1].
Therefore, SustainCERT is fully eligible and authorized to carry out validation functions, including those triggered by a change in applied methodology.
The VVB has reviewed the revised PD and confirms that the project has been assessed against all applicable requirements of VM0050 v1.0, and the findings of this assessment are reflected in the Verification Report.

[1] https://verra.org/validation-verification/sustaincert-sa/

3 VALIDATION FINDINGS

During this verification (1st verification) for the group project, the GP underwent a methodology change, the group project activity was earlier registered with methodology VMR0006: Methodology for installation of High Efficiency Firewood Cookstoves v1.2. However, now the project is taking a deviation and changing the methodology to VM0050 Energy Efficiency and Fuel-Switch Measures in Cookstoves V 1.0. The following section of the report mentions the detailed assessment of the validation activities performed as a part of verification of the group project.

3.1 Methodology Deviations

There is a methodological deviation that has been applied in the project inline with clarification 6 of the project.

The project proponent has utilized baseline survey data collected in 2022 to calculate average adult equivalent values. The methodology VM0050 v1.0, clarification 6 released on Feb 2025 prescribes a detailed categorization by gender and age for determining the adult equivalent. PP has applied the value inline with census data and PNG population statistics (https://www.nso.gov.pg/statistics/population/) which demonstrate the male and female population nearly same.

Household profile (VM0050 binding questionnaire)	Value to be used for calculating Adult Equivalent	Household profile (baseline survey)	Value used for calculating Adult Equivalent
Females, over 14 years	0.8	Adults, 16 -63 years	0.9
Males, 15-59 years	1.0		
Males, over 59 years	0.8	Adults, 64 years and above	0.8
Children, 0-14 years	0.5	Children, under 15 years	0.5

Based on the cross check from the census data and 2021 National Statistical Office (NSO) census data for Papua New Guinea, above calculation is found to be conservative. The data was compared with recent field measurements conducted, and it was observed the variation less the 1%, demonstrating above method to be more conservative. Therefore, inline VCS standard, section 3.20, considering the conservativeness of deviation and government approved data sources taken it is accepted by VVB.

3.2 Project Description Deviations

In line with the Procedure to change Methodology through a Project Description deviation V4.0, section 2.3, point 2.3.2, the assessment team has conducted the assessment of the project description document with methodology, VM0050 Energy Efficiency and Fuel-Switch Measures in Cookstoves V 1.0. The detailed assessment of the relevant sections can be found below.

A) Project Eligibility:

Exclusion under table 2.1 of VCS standard: The group project activity is distribution of improved cookstoves in the Papua New Guinea. The ICS distributed are energy efficient, reducing GHG emission. The project does not involve any activity that is explicitly excluded under Table 2.1 of the VCS Standard, such as grid-connected electricity generation, waste heat recovery, fossil fuel switching, or hydrofluorocarbon (HFC-23) emission reduction. Thus, the project is not excluded under Table 2.1 and meets the scope requirement of VCS standard v4.7.

The project meets the requirements of pipeline listing[/27/]: The pipeline listing date was 14/07/2023, preceding the project start date of 23/09/2024, ensuring compliance with standardized additionality requirements. The opening meeting with the first Verifier took place on 05/05/2023, followed by the VCS Registration request on 09/11/2023. The project was formally registered on 14/06/2024 and is scheduled for further verification. These steps align with the VCS Standard v4.7 Section 3.8.1 and section 4.1.5, ensuring that the validation and registration processes are conducted within the prescribed timelines. Thus, the requirement is met. CL 01 was raised and closed successfully.

Methodology eligibility capacity limits: The project applies VM0050: Energy Efficiency and Fuel-Switch Measures in Cookstoves, v1.0, an approved methodology under the Verra Standard. The methodology is applicable to the project activity, the applicability criteria of the methodology have been verified below. The methodology does not introduce any scale or capacity limits. Additionally, double counting prevention measures have been implemented through the assignment of a Unique Serial Number (USN) for each ICS, recorded in an electronic database[/28/]. This system ensures that emission reductions are not claimed multiple times. CL 07 was raised and closed after reviewing the sufficient evidence. The details related to the unique Number have been verified and assessed in detail and also the implementation of the process has been checked on site. Thus, the criteria have been met.

Therefore, it has been concluded that the project activity is eligible under the VCS program.

B) Project start date and crediting period:

The project start date has been changed to 23/09/2024, based on the first ICS distributed under the project activity with USN number as FDYP55[/21/], the assessment team has verified the end user contract for the 1st ICS distributed and also it was verified with PP during the onsite visit for the project, the start date mentioned for the project activity is inline with VCS standard V4.7, section 3.8. The earlier date mentioned in the registered PD is 01/01/2024 which was the estimated date. Therefore, the date 23/09/2024 is accepted as the starting date based on the evidence provided. The crediting period for the project is 23/09/2024 to 22/09/2034 Thus, the start date and crediting period of the project is accepted.

C) Project scale and estimate emission reductions:

The group project is calculating emission reductions based on VM0050: Energy Efficiency and Fuel-Switch Measures in Cookstoves, v1.0, and is categorizing the project as large scale inline with VCS standard, section 3.10, “Large projects: Greater than 300,000 tonnes of CO2e per year”.
Based on the assumption values taken by the project to calculate the estimated emission reduction it can be concluded that the project is inline with the methodology and will be able to achieve the emission reductions mentioned.

D) Conditions prior to project initiation

The conditions prior to the project initiation are same as described in the baseline scenario for the project, continued use of non-renewable wood fuel using unimproved open fire three stone firewood and the already validated statement remain valid during the methodology change as well. More details related to the crosscheck of baseline scenario can be found in section 2.3 of the report.

E) Title and reference of methodology

During the current Monitoring the project undergoing a methodology change from VMR0006 Methodology for installation of High Efficiency Firewood Cookstoves v1.2 to VM0050 Energy Efficiency and Fuel-Switch Measures in Cookstoves, 1.0 inline with the guidelines release by VERRA on 9/10/2024 (https://verra.org/verra-releases-new-cookstoves-methodology/).

Along with the methodology project applies the latest version of tool 30 Calculation of the fraction of non-renewable biomass V4.0 and CDM Guideline: Sampling and surveys for CDM project activities and programs of activities, v4.0 and CDM sampling standard, V9.0, all the applicable tools and methodology applies the latest version therefore acceptable.

F) Applicability of methodology

The project activity currently applies methodology VM0050 Energy Efficiency and Fuel-Switch Measures in Cookstoves, 1.0. The table below assesses the applicability conditions of methodology:

S. No	Applicability Condition	PP's Justification	Assessment opinion
1.	The project activity corresponds to: a) Replacement of non-renewable biomass (e.g., firewood, charcoal)-fired cookstoves with any of the following: i) More efficient project devices that use the same fuel as in the baseline; ii) Efficient project devices fired by renewable biomass or bioethanol; iii) Efficient project devices fired by liquefied petroleum gas (LPG); or iv) Electric-powered project devices.	The project activity involves the replacement of non-renewable biomass with more efficient project devices that use the same fuel as in the baseline (firewood), therefore this criterion is met.	The group project activity is located in the PNG and is aimed at distribution of more efficient cookstoves with the same fuel as in the baseline. The information related to this have been verified during the validation and during the current site visit that the fuel for the project remains the same as baseline. However, the stove has been improved. Therefore, the criteria is found to be met.

	b) Replacement of solid or liquid fossil fuel (e.g., coal, kerosene)-fired cookstoves with any of the following: i) Efficient project devices fired by renewable biomass or bioethanol; ii) Efficient project devices fired by LPG; or iii) Electric-powered project devices.		
2.	Project devices are used in households, communities, institutions, or small or medium enterprises (SMEs),4 collectively referred to in this methodology as the "target population."	The project activity will be used in an end-user's household for heating and cooking purposes. Therefore, this criterion is met	In accordance with the site visit conducted and the project database submitted/16/, it has been verified that the project activity aims at distributing the ICS to the end users households. Therefore, the condition is found to be met.
3.	Where renewable biomass is used, it is exclusively renewable and qualifies as one of the following: a) A by-product, residue, or waste stream from agriculture, forestry, and related industries; or b) Originating from dedicated plantations that comply with all relevant applicability conditions in the most recent version of CDM TOOL16	The project activity does not use renewable biomass therefore this criterion does not apply	The project activity aims at distribution of the improved ICS. However, the fuel remains the same, continued used of non-renewable biomass as in the baseline, it has been verified form project validation as well as on site visit conducted during this monitoring period. Therefore, the criteria does not apply.

4.	Where biomass residues are used, they would have been left to decay or burned without energy recovery before implementation of the project activity, and their use does not involve a decrease in carbon pools – in particular of dead wood, litter, or soil organic carbon – on the land areas from which the biomass residues originate.	The project activity does not use biomass residues therefore this criterion does not apply.	The project activity aims at distribution of the improved ICS. However, the fuel remains the same, continued used of non-renewable biomass as in the baseline, it has been verified form project validation as well as on site visit conducted during this monitoring period. It has been verified that the project activity does not use biomass residue. Therefore, the criteria is not applicable.
5	Where biomass residues from a production process are used, project implementation does not result in an increase in the processing capacity of raw input or any other substantial changes (e.g., product change) in this process.	The project activity does not use biomass residues therefore this criterion does not apply.	The project activity aims at distribution of the improved ICS. However, the fuel remains the same, continued used of non-renewable biomass as in the baseline, it has been verified form project validation as well as on site visit conducted during this monitoring period. It has been verified that the project activity does not use biomass residue. Therefore, the criteria is not applicable.

6	Where more than one type of renewable biomass is used, each of the biomass types used complies with the applicability conditions.	The project activity does not use renewable biomass therefore this criterion does not apply.	The project activity aims at distribution of the improved ICS however, the fuel remains the same, continued used of non-renewable biomass as in the baseline, it has been verified form project validation as well as on site visit conducted during this monitoring period. Therefore, the criteria does not apply.
7	Where project activities introduce renewable biomass as charcoal, it is renewable charcoal produced by efficient charcoal production processes (e.g., retort sedentary kilns, improved sedentary kilns, Casamance kilns). Methane produced during the charcoaling process is captured and destroyed or combusted for energy purposes.	The project activity does not introduce renewable biomass such as charcoal into the project boundary therefore this criterion does not apply.	The project activity aims at distribution of the improved ICS however, the fuel remains the same, continued used of non-renewable biomass as in the baseline, it has been verified form project validation as well as on site visit conducted during this monitoring period. It has been verified that the project activity does not use charcoal biomass residue. Therefore, the criteria is not applicable.
8.	Project devices using renewable biomass (fuel-switch) or non-renewable biomass (improved efficiency) are single-pot,	The project activity uses non-renewable biomass single-pot portable cookstoves that has an initial thermal efficiency	PP has submitted the thermal efficiency test laboratory results, demonstrating that efficiency of the stove

	multi-pot portable, or in-situ cookstoves with an initial thermal efficiency8 of at least 25%.	of 52%, therefore this criterion is met.	is higher than 25%[/18/]. Therefore, the condition is found to be met.
9	Project devices using LPG or bioethanol are single-pot, multi-pot portable, or in-situ cookstoves with an initial thermal efficiency of at least 30%	The project activity uses ICSs that utilize firewood as a fuel and does not involve LPG or bioethanol therefore this criterion does not apply.	The project activity uses on non-renewable biomass as a fuel source. Therefore, the condition is not applicable.
10	Electric project devices meet the maximum risk factor score of 15 on the Cookstove Durability Protocol and have the following minimum thermal efficiency: a) Hot plates and electric hobs: 40% Induction stoves and other electric stoves: 70%	The project activity uses ICSs that utilize firewood as a fuel for cooking and heating purposes, therefore this criterion does not apply.	The project activity uses on non-renewable biomass (firewood) as a fuel source. Therefore, the condition is not applicable.
11	Project devices using LPG comply with all of the following conditions: a) The baseline fuel either includes non-renewable biomass or is a more carbon intensive fossil fuel (demonstrated by the baseline survey, see Section 6.2); b) The project has a provision for metering LPG supplied to each consumer at the LPG filling station, in order to determine household LPG consumption; and c) The	The project activity uses ICS that utilize firewood as a fuel for cooking and heating purposes, therefore this criterion does not apply	The project activity uses on non-renewable biomass (firewood) as a fuel source. Therefore, the condition is not applicable.

	project does not seek to issue any carbon credits for periods after 31 December 2045		
12	Electric project devices use the following electricity sources: a) Decentralized renewable energy systems: Decentralized energy systems using fossil fuels are not eligible, except for backup generators that supply less than 1% of the annual electricity of the decentralized renewable energy system.12 b) Self-generated renewable electricity (with a maximum of 20% electricity from non-renewable sources for backup); or c) National or regional electricity grid.	The project activity uses ICS that utilize firewood as a fuel for cooking and heating purposes, therefore this criterion does not apply	The project activity uses on non-renewable biomass (firewood) as a fuel source. Therefore, the condition is not applicable.
13	The project proponent designs incentive mechanisms to reduce the use of inefficient baseline devices and practices that can be replaced by the project devices and describes these mechanisms in the project description	The project proponent employs the following incentive methods to reduce the use of inefficient baseline devices and practices: <u>Household Education and Encouragement:</u> The project proponent, through cookstove ambassadors, routinely visits end-users throughout the project lifetime to collect feedback, maintain the project cookstoves, and address questions or	. It has been confirmed that: Routine site visits by the project proponent to end-user households to collect feedback, address user queries, and provide ongoing support for the correct usage and maintenance of improved cookstove units. Engagement of

		queries regarding the project activity to encourage the continued use of the project cookstoves. <u>Behavioural Change Campaigns:</u> Awareness champaigns are also held by field assistants and cookstove ambassadors in the form of community meetings featuring cooking demonstrations, peer testimonials within villages to promote and educate households on the health, environmental, and economic benefits of transitioning to improved cookstoves, thereby motivating voluntary behaviour change. Additionally, the project proponent works with the Western Pacific University (WPU) of Papua New Guinea to develop and train locals on project management and carbon knowledge. Graduates of this course would be able to work with the project proponent formally during the project lifetime. <u>Monitoring and Enforcement:</u>	Cookstove Ambassadors to foster continuous awareness of the benefits of ICSs over traditional open-fire cooking. Training and employment of local community members, verified through employment and training records reviewed by the VVB, ensuring local capacity building and ongoing project implementation support. Additionally, the project employs community-based incentive strategies: Quarterly monitoring of ICS adoption, with results tracked and reviewed by the PP. A 'Cook-Off Event' incentive, awarded to villagers demonstrating the most significant progress in adopting ICSs, thereby encouraging positive behavioral change and sustained usage.

		During the project lifetime, visits done for household education and encouragement will also monitor the usage rate of the ICS. This effectiveness of the abovementioned incentive mechanisms will be monitored as a decrease in stove stacking factor. Hence, this criterion is met	The CL 16 was raised and resolved successfully. Based on the evidences reviewed, the assessment team confirms that project meets the criteria. Therefore, the criteria is found to be met.
14	Where a project device has ended its technical life, the project proponent either replaces it with a comparable or better project device or retrofits its essential components to continue meeting the minimum service level requirements (i.e., thermal energy generation), otherwise no further emission reductions may be claimed for the project device	Where the ICS has ended its technical life, the project proponent does not replace the ICS. Instead, no further emission reductions will be claimed for the ICS, therefore this criterion is met.	In accordance with the discussion with PP it has been verified that PP will not claim any emission reductions for the stoves that fails to meet the eligibility criteria of technical lifetime. Therefore, the criteria is found to be met.
15	Project proponents implement a method for the distribution and identification of project devices that avoids double counting of emission reductions by other mitigation actions and includes unique product identification on the stove	In order to avoid double counting emission reductions through the method of distribution, each ICS will carry a USN and the distribution of each device is logged into an electronic database which includes unique information for	Each improved cookstove distributed under the project is assigned a Unique Serial Number (USN), which is physically marked on the device. These USNs are recorded in an electronic database

	itself at the time of distribution/sale (e.g., program logo, alpha/numeric ID, and end-user location, such as geographic coordinates, complete address information).	each recipient (i.e., Unique Serial Number, geographic coordinates, etc.) upon receipt of the device. The electronic database combined with internal checks done by the PP will ensure that no USNs are double counted. During the monitoring period, survey samples will be only taken from the USNs that are in the electronic database of end-users eliminating the risk of double counting from any other projects within the same geographical location.	that also captures household-level details including geographic coordinates, full address, name of recipient, distribution date, and project staff involved. The distribution is further validated through photographic evidence and End User Agreements. Internal validation checks are routinely conducted by the project proponent to ensure there are no duplicate entries or overlapping USNs. Furthermore, survey samples during the monitoring period were drawn exclusively from the verified USN pool in the central database, ensuring alignment with the documented population and avoiding overlap with other projects in the same region. CL 07 was raised and closed successfully. Therefore, this criteria is found to be met.
16	The project complies with any national, sub-national, or local regulations or guidance for the	The project activity complies with any national, sub-national, or local regulations or	PP has provided the Signed documents related to NOLs related from the

	installation, commercialization, distribution, and use of improved cookstoves and/or fuel supply and use for the target population. National, regional, and local regulatory frameworks for the provision of the type of thermal energy services provided by the project activity must be documented.	guidance for the installation, commercialization, distribution, and use of improved cookstoves and/or fuel supply and use for the target population. At every stage of project inception, the project proponent has been in touch with all levels of relevant authorities and have gained approval, therefore this criterion is met.	national authorities and local authorities[/11/]. These documents have been checked in detail, along with this project stakeholder consultation records have also been checked, alongside this it has been checked that there has been no national regulation related to mandatory cookstove distribution. therefore, it has been concluded the criteria is found to be met.
17	Where project activities reduce emissions from non-renewable biomass, including firewood and charcoal, the risk of double counting is assessed on a national basis by evaluating at validation and crediting period renewal whether there are REDD+ projects or jurisdictional REDD+ programs whose project boundary overlaps with the expected fuel source area of the project. The project proponent must report on the findings of this assessment for informational purposes in the project description.	Currently, there are no REDD+ projects or jurisdictional REDD+ programs with the boundary of PNG Highlands, therefor this criterion is applicable.	The assessment team has check the status of the project active PNG[/24/], It has been found that there are no active projects REDD+ project in PNG, therefore, the criteria is found to be met.

In accordance with above assessment, it is concluded that methodology is rightly applied.

G) Project boundary

Project determines the boundary inline, VM0050 Energy Efficiency and Fuel-Switch Measures in Cookstoves, V1.0, In line with methodology section 5, project boundary includes the project devices, the geographical site where they are located, and the locations from which baseline and project fuels are sourced.

The assessment team was able to ensure that the approved methodology addresses all the identified emission sources that are impacted by the project activity as mentioned in section 3.3 of revised PD.

H) Baseline scenario

The baseline scenario of the projects remains the same as validated during the initial validation of the project. The baseline scenario remains valid inline with section 6 of the applied methodology VM0050: Energy Efficiency and Fuel-Switch Measures in Cookstoves, v1.0. The baseline scenario defined is continued use of inefficient cooking stove in the project area with high dependency on non renewable woody biomass.
PP has addressed all the 3 steps required to assess the baseline scenario inline with the updated methodology:
Step 1: Identify the alternative baseline scenarios:
Baseline of the area remains to be continued use of non-renewable biomass in the project area, which was demonstrated by the baseline survey conducted by the PP with third party in November 2022, the study was again reviewed by the SPFEC, a letter was sent on 15th Jan 2025, confirming the baseline scenario. Additionally research[2] study were also verified to substantiate the above.
Step 2: Considering existing and forthcoming government policies and legal requirement
It was verified through the independent research as well that there are no government policies regarding the cooking technology.
Step 3:Assess financial, institutional and information barriers
It was verified through independent research that no such project has been implemented in the similar area, and no financial institute is supporting the project and also there is lack of awareness in the people which will be covered by awareness campaigns by PP.
Thus, all the 3 steps were successfully met.
The project proponent (PP) has adhered to the questionnaire provided in Appendix 1 of the methodology. Baseline KPTs were also conducted to verify baseline fuel consumption, which was subsequently validated through a letter from SPFEC dated January 2025.
The PP has taken a deviation related to the calculation of Adult Equivalent; however, the calculation was found to be conservative. Further details are provided in Section 3.1 of the report.
Controlled households: As per methodology, "These control households must be established prior to validation and must be shown to be statistically equivalent to the pre-project conditions of project households regarding baseline fuel type(s) and percentage of use by the target population, source(s) of each baseline fuel, and baseline technologies for cooking. For existing projects updating to VM0050, control households must be established before the first verification under VM0050".
In line with this, the PP has identified control households and areas within the Western Highlands region (KML files have been submitted). However, registration of these households has not been possible due to on-ground security concerns. It has been observed that residents in those areas are aware of the project and stove distribution due to awareness campaigns and prior outreach. Registering households in these areas and subsequently not providing stoves could pose a security risk to project personnel as well as to end users in adjacent villages. Which can also been with security issues already existing in PNG (https://www.gov.uk/foreign-travel-advice/papua-new-guinea/safety-and-security).

Koalilombo Ward has been designated as the control area for the project. It is located within the Western Highlands region and, based on available research, exhibits similar cooking and meal preparation habits. Therefore, it is considered that the baseline for this area remains consistent with that determined during the initial verification. The assessment team has reviewed and verified the submitted KML files and supporting research for the region.

As per the methodology section 6.2, “The initial baseline survey must be performed prior to validation. The project proponent may employ local third-party agencies to carry out the baseline survey. Follow-up baseline surveys must be conducted every two years in control households that do not participate in the project.” In line with Clarification related to methodology, a follow-up survey will be conducted every five years.

Thus, during the methodology change there is no change in the baseline scenario for the project. The baseline scenario defined is also appropriate inline with the VCS standard 3.13, and through independent research it has been verified that there are no laws and regulations in PNG related to the project activity such as minimum product efficiency standards, air quality requirements, carbon taxes, and subsidies. Therefore, the baseline scenario for the project is applicable. CL2 was raised for the same and closed successfully.

I) Additionality

The group project has demonstrated additionality inline with the methodology VM0050 Energy Efficiency and Fuel-Switch Measures in Cookstoves, V1.0,

1. Regulatory surplus: the project activity is located in PNG, which is included in annex 1 country, And it has been verified with the help of independent research[3] that there is no low or regulation enforced which mandates the project activity.
2. Positive list: Project meets the eligibility criteria set out in the methodology. It has been conformed with the help of end users interview and end user agreement, that project ICS has been given to the end users free of cost and there no other sources of revenue.

Therefore, in line with the applied methodology project technology is deemed additional by meeting step 1 and step 2 of the applied methodology.

J) **Quantification of GHG emission reductions and carbon dioxide removals**

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[2] ACIAR (2013). Promoting diverse fuelwood systems in Papua New Guinea- Final Report.pdf
Hanson etal 2001_PNG Rural development handbook_IMPORTANT.pdf
Murphy 2009_A critique of fuelwood in the rural households in Eastern Highlands of PNG.pdf
Nuberg 2015_Developing the fuelwood economy of Papua New Guinea.pdf

[3] https://faolex.fao.org/docs/pdf/png51966.pdf
food-safety-and-public-health

The project activity is quantifying the emission reductions based on methodology VM0050: Energy Efficiency and Fuel-Switch Measures in Cookstoves, v1.0, the assessment team has assessed the details it has been observed that it conservatively estimated. Further details related to the calculation can be found in section 4.4 of this report.

Leakage assessment: The PD has assessed the additionality on the basis of methodology VM0050: Energy Efficiency and Fuel-Switch Measures in Cookstoves, v1.0, In line with section 8.3, based on the based with association with reduced or avoiding the use of non-renewable biomass, the leakage gas been correctly applied as 0.95. There it is acceptable.

K) Methodology deviations

Methodology deviations are described in section 3.1 of the report. **Monitoring plan**

The assessment team reviewed the monitoring plan described in the Project Design Document and confirms that it clearly identifies all relevant parameters to be monitored as per the applied methodology VM0050: Energy Efficiency and Fuel-Switch Measures in Cookstoves, Version 1.0.
Key monitored parameters include (more details related the parameter can be found in section 4.3 and 4.4 of the project report):
- Number of stoves distributed and operational
- Thermal efficiency of project devices
- Fraction of non-renewable biomass (fNRB)
- Usage rate of the improved cookstoves
- Baseline and project scenario fuel consumption
-Leakage emissions (via applied adjustment factor)

It has been verified by the assessment team that monitoring procedures outlined in the PDD ensure traceability, consistency, and conservativeness based on the onsite visit and the evidence submitted. The project utilizes a digital data management system for stove registration, tracking each stove by a Unique Serial Number (USN) along with GPS coordinates and household-specific information[/28/]. The procedures for data collection, entry, validation, and storage are robust and compliant with the methodology requirements.

Monitoring equipment and methods, including household surveys, field verification, and sampling protocols, are designed in accordance with the methodological guidance. Internal quality assurance protocols and field team training further enhance data credibility.

Based on document review and site audit findings, the assessment team concludes that the monitoring plan has been effectively implemented and is fully in adherence with the requirements of the applied methodology and the referenced tools. No deviations were noted, and the system is capable of ensuring accurate and consistent monitoring throughout the crediting period.

Thus, it has been concluded that the project activity meets all the requirements mentioned in the section 2.3.2 of procedure to change methodology through project description deviation. Therefore, the assessment team concludes that the project activity is eligible to apply the methodology VM0050, V1.0.

3.3 New Project Activity Instances in Grouped Projects

The assessment team reviewed the procedures and evidence submitted to validate the inclusion of new project activity instances under the grouped project activity, During this monitoring period, PP has added 19,206 PAI under the group project activity. Table below mentions the assessment criteria for each of the conditions specified:

Criteria	Applicability	Assessment opinion
Meets the applicability conditions set out in the methodology applied to the project	New PIs will meet the applicability conditions set out in Section 4 of the methodology	All the new ICS distributed are within the boundary of PNG as verified from distribution database and KML files and meet the efficiency criteria of 25%. The detail assessment of methodology applicability criteria can be found in section 3.1 of the report.

Use the technologies or measures specified in the project description.	Only ICSs that conform with the grouped project description are to be distributed in the project. The ICS will be chosen to deliver a level of service at least equivalent to the baseline appliance.	The assessment team reviewed the stove models distributed under the grouped project from the distribution database and also sample households were visited during onsite visit[/12/]. Only Burn-manufactured[/15/] ICS units, which are explicitly listed and approved in the project description, were distributed to households. These stoves feature a stainless-steel insulated combustion chamber and exceed the minimum thermal efficiency of 25%, as required by the applied methodology VM0050[/08/]. Therefore, the criteria are found to be met.
Applies the technologies or measures in the same manner as specified in the project description.	ICSs distributed in the PIs will adhere to the grouped project description	The assessment team has verified the distribution database[/16/], and also during onsite visit it has been confirmed that project has been distributed as per the description in the PDD and the same stive model has been distributed throughout. Therefore, the condition is found to be met.

Are subjected to the baseline scenario determined in the project description for the specified project activity and geographic area.	New PIs will be installed only in regions within the geographic borders of PNG subject to the same baseline scenario determined in Section 3.4 in this Project Description	The assessment team has gone through the monitoring survey report/16/ and conducted onsite visit to verify the baseline. Additionally, as during this verification Project is Appling the procedure to change the project methodology, interview with the stakeholder were conducted (detail of which can be found in section 2.3 of the report) to verify the baseline, and it was concluded that baseline is correctly defined. Therefore, the criteria are found to be met.

Have characteristics with respect to additionality that are consistent with the initial instances for the specified project activity and geographic area. For example, the new project activity instances have financial, technical and/or other parameters (such as the size/scale of the instances) consistent with the initial instances, or face the same investment, technological and/or other barriers as the initial instances.	All new PIs will use the activity method for demonstration of additionality. Step 1: Regulatory Surplus There is no government mandated programme or policy in host country of this project ensuring the distribution of new project activity instances. Step 2: Positive List The inclusion of new project activity instances will comply with the positive list as it satisfies criterion 1 where it meets all the applicability conditions of the methodology.	The assessment team has verified the project detail for additional PAI in line with the criteria mentioned Regulatory surplus: It has been confirmed that here is no government-mandated policy or regulation in Papua New Guinea requiring the distribution or adoption of improved cookstoves such as those implemented under the project. PP has taken all the required permissions form government/22/. CL02 was raised related to can be checked in appendix 4 of the report. The assessment team has reviewed the eligibility of new project instances under the positive list criteria. It was confirmed that the new PIs conform to the conditions outlined in the methodology and share the same characteristics such as scale, technology, and financial/investment barriers as the initial instances. Therefore, the criteria are found to be met.

Occur within one of the designated geographic areas specified in the project description.	New PIs will only occur within the boundary of PNG.	The assessment team has verified that distribution database and also conducted onsite visit (more detail in section 2.3 of the report) it has been verified that the project is implemented within the project boundary, therefore the criteria is found to be met.
Comply with at least one complete set of eligibility criteria for the inclusion of new project activity instances. Partial compliance with multiple sets of eligibility criteria is insufficient.	New PIs will comply with at least one complete set of eligibility criteria for the inclusion into the GPA.	The assessment team has reviewed the inclusion of new project activity instances under the grouped project activity and assessed their compliance with the applicable eligibility requirements as per the applied methodology VM0050, V1.0. The Monitoring Report confirms that each new PI complies with at least one complete set of eligibility criteria

Be included in the monitoring report with sufficient technical, financial, geographic, and other relevant information to demonstrate compliance with the applicable set of eligibility criteria and enable sampling by the validation/verification body.	New PIs will demonstrate compliance with the applicable set of eligibility criteria and enable sampling by the validation/verification body through the inclusion of sufficient technical, financial, geographic, and other relevant information within the monitoring report.	The assessment team confirms that the Monitoring Report provides sufficient technical, financial, geographic, and other relevant information for the newly included PIs. The information provided has also facilitated the sampling and assessment of the newly added instances during the verification process. Based on this review, the verification team concludes that the inclusion of new PIs has been carried out in accordance with the eligibility criteria specified in the methodology, and that all relevant supporting documentation and evidence were adequate for verification. Therefore, the criteria are met.
Be validated at the time of verification against the applicable set of eligibility criteria.	New PIs will be validated at the time of verification against the applicable set of eligibility criteria.	The assessment team has verified the distribution database, monitoring survey and has conducted onsite visit, also cross verifying the emission reduction calculation sheet, it has been verified that they all the PAI added during verification meets the eligibility criteria.

Have evidence of project ownership, in respect of each project activity instance, held by the project proponent from the respective start date of each project activity instance (i.e., the date upon which the project activity instance began reducing or removing GHG emissions).	New PIs will provide evidence of project ownership, in respect of each project activity instance, held by the project proponent from the respective start date of each project activity instance.	The assessment team has verified the end user agreements for the end users on the sampled basis as well ownership certificate from burn has also been verified, therefore, it has been confirmed that PP has the ownership of the GHG emission reductions.
Have a start date that is the same as or later than the grouped project start date.	The start date of new PIs will be after 23/09/2024, after the start date of the GPA.	The assessment team has verified the start date if the project activity as 23/09/2024 by the end user agreement/21/ also confirmed during the PP interview therefore, the criteria is met.
Be eligible for crediting from the start date of the instance through to the end of the project crediting period (only). Note that where a new project activity instance starts in a previous verification period, no credit may be sought for GHG emission reductions or removals generated during a previous verification period (as set out in Section 3.4.4) and new instances are eligible for crediting from the start of the next verification period.	New PIs will be eligible for crediting from the start date of the instance through to the end of the project crediting period (only). For example, when a PI starts with a fixed crediting period of 10 years (fixed), the project device/stove must be proven at inclusion that it will be able to sustain through the crediting period. This may be proven through the Project Device Lifespan Specification document.	With respect to crediting eligibility, the assessment team confirms that the new PIs are eligible to generate emission reductions only from the start date of their inclusion and through to the end of the project crediting period. No emission reductions have been claimed for any period prior to their inclusion.
Not be or have been enrolled in another VCS project.	New PIs must declare that it has not or have been	The assessment team confirms that all newly included project instances have not been part of any

	enrolled in another VCS project.	other project, all project currently in PNG have been Checked[/24/] it has been confirmed that there are no similar projects in the same are additionally, PP has submitted a formal declaration stating that they have not been previously enrolled in any other VCS project activity. This declaration along with the assessment conducted bt the assessment ensures compliance with Verra's requirements to prevent double counting or double issuance of emission reductions.
Adhere to the clustering and capacity limit requirements for multiple project activity instances set out in 3.6.8 - 3.6.9 of the VCS Standard.	New PIs will adhere to the clustering and capacity limit requirements set out in Section 3.6.8 – 3.6.9 whereby the Project participant (PP) shall include all project activity instances within ten kilometres of another instance of the same project activity and with the same project proponent and where a capacity limit applies to a project activity included in the project, no project activity instance shall exceed such limit.	PP has provided the GPS coordinate and KML files, The assessment team has assessed details, it has been confirmed the details are appropriate. CL09 was raised and was closed after receiving satisfactory response and evidence from PP.

Based on the above assessment, it has been verified that all the new PAI added in the project meets the eligibility criteria of inclusion of the new PAI inline with VCS standard requirement, Therefore, a reasonable level of assurance is achieved.

3.4 Baseline Reassessment

Did the project undergo baseline reassessment during the monitoring period?

☐ Yes ☒ No

The project is undergoing methodology change from VMR0006 V1.2 to VM0050 V1.0 as well during this verification, The baseline was assessed against the new methodology and it has been verified that the original baseline remains valid. Section 2.3 demonstrates the interviews conducted to verify the original baseline onsite and it has been confirmed that original baseline remains valid and thus, the reasonable level of assurance has been met.

4 VERIFICATION FINDINGS

4.1 Project Details

The project “Improved Cookstove Grouped Project in Papua New Guinea” (VCS ID 4416) falls under sectoral scope 03 – Energy Demand, as appropriately identified in the registered and new VCS Project Description and Monitoring Report. The project is implemented within the national boundary of Papua New Guinea, with all activities taking place in the Southern Highlands Province during the monitoring period as verified through the project database and also onsite visit (more detail in section 2.3 and 2.4 of the report) was conducted, sample households were visited to confirm the Implementation.
The project proponent is TEM PNG Cookstoves Project Pte. Ltd., which is correctly listed in the VCS PD. No other entities are involved in the project implementation. This was verified against the information published on the Verra Registry

The start date for the grouped project is 23/09/2024 which is the date of installation of first stove in the grouped project[21/]. Assessment team would like to mention here that the project proponent has considered each ICS distributed as a project activity instance. The selected crediting period for the grouped project activity is a fixed 10-year period, commencing on 23/092024 and ending on 22/09/2034, which is consistent with the project start date and conforms with the provisions of the VCS Standard (v4.7). The project is correctly classified as a large project as per registered PD and VCS Standard[/1/] as the average annual emission reductions are more than 300,000 tCO2e. Project design, including eligibility criteria for grouped projects are assessed in section 3.4 of the report.

The grouped project activity envisages implementation of Burn cookstove[/15/] model “a stainless-steel insulated combustion chamber, with a high thermal combustion efficiency (over 25%)” which is verified during the site visit[/12/] and as well as the verified from the manufacturer technical specification and laboratory test[/18/] provided by the project proponent.

Item	Evidence gathering activities, evidence checked, and assessment conclusion:
Audit history	The project activity is undergoing 1st Verification during its fixed 10-year crediting period. The assessment team has verified that audit history using the Project webpage/4/, the project has been registered under VERRA. Validation – VCS – Dated 24/04/2024 Verification – VCS – currently undergoing – 23/09/2024 to 31/12/2024 It has been concluded that the information presented in section in 1.2 of MR is appropriate. Link to project webpage - *https://registry.verra.org/app/projectDetail/VCS/4416*
Double counting and participation under other GHG programs	The project activity is not registered under any other GHG and non-GHG program or registry. This has been confirmed by assessment team by means of declaration submitted by PP (#CL3) and research on relevant applicable registries and available information in public domain and through declaration from the Project Owner. The assessment team has searched for similar projects having same nature, project technology, location and project proponent as well as legal owner. It was concluded that no such projects having same location and geo-coordinates, technology or project/legal owners are registered in various carbon schemes like CDM, GCC, and Gold Standard.
No double claiming with emissions trading programs or binding emission limits	The project activity has been thoroughly searched on the various websites of various emission trading programs and or ETS programs running in various counties, using nature, project technology, location and project proponent as well as legal owner as the search option.

No double claiming with other forms of environmental credit	As discussed above the, the project has been thoroughly searched in the currently active environment schemes and found that the project is not registered with any other environmental credits. PP has also submitted a declaration stating that the project is not registered with any other registry (refer CL#3 in appendix-4 of this report). Therefore, it can be concluded that the project is not claiming any credits from other programs/schemes.
Supply chain (scope 3) emissions double claiming	The grouped project activity which aims at distribution of ICS to households does not result in any Scope 3 emissions in its supply chain.
Sustainable development contributions	As claimed in section 1.12 of the monitoring report, the project is claiming positive impact for the following: SDG 1 (No Poverty): Reduction in household expenditure through decreased reliance on purchased firewood, enabled by the distribution of 19,206 fuel-efficient improved cookstoves (ICSs). SDG 3 (Good Health and Well-being): Reduced indoor air pollution due to an 81% decrease in PM2.5 emissions, improving respiratory health conditions for ICS users. Verified using monitoring survey and Laboratory testing reports reports. SDG 4 (Quality Education): Non-formal vocational training provided to 24 local staff, including both operational and monitoring roles, enhancing community skill levels. Verified using training records/17/. SDG 5 (Gender Equality): Engagement of women in project implementation, with 37.5% of the trained workforce being female during this monitoring period. Verified using employment records/14/ SDG 7 (Affordable and Clean Energy): Increased access to clean and efficient cooking technology through widespread ICS deployment. Verified using distribution database/16/

	SDG 8 (Decent Work and Economic Growth): Creation of 24 local jobs (2 full-time, 22 part-time), with fair compensation and ongoing capacity-building activities. Verified using employment records/14/ SDG 13 (Climate Action): Total GHG emissions reductions of 13,443tCO_2e during the monitoring period, directly attributable to reduced fuelwood usage. SDG 15 (Life on Land): An estimated 13,443 tonnes of non-renewable biomass saved during the monitoring period, supporting forest conservation efforts. Verified from the survey records and emission reduction calculation sheet. The reported SDG impacts were supported by quantitative data (e.g. fuelwood savings, emission reductions, training numbers), field monitoring results, and documented stakeholder consultations. The assessment team reviewed these claims against the Monitoring Report, sampling records, and supporting documentation, and found the information to be consistent and credible. Based on this review, the assessment team concludes that the reported SDG contributions are valid, appropriately monitored, and substantiated for the current monitoring period. CL10 is raised by assessment team in regard to SDG claim for this monitoring period and closed successfully.
Additional information relevant to the project	No commercially sensitive information has been excluded by the PP.

4.2 Safeguards and Stakeholder Engagement

4.2.1 Stakeholder Identification

Stakeholder Identification	Evidence was gathered through review of the Monitoring Report (Section 2.1.1), stakeholder consultation summaries, stakeholder invitation and field visit interviews/13/. Stakeholders identified include household end-users, community leaders, youth leaders, local stove distributors, and representatives of women's groups. The process of identification follows the approach established at validation and was confirmed during the site visit. The verification team concludes that stakeholder identification remains valid and comprehensive.
Legal or customary tenure/access rights	The project activity does not involve land use, land access, or biomass harvesting rights. Cookstoves are distributed for voluntary use within individual households. No changes to legal or customary tenure or access rights were observed or reported. The verification team concludes that this requirement is not applicable to this project.
Stakeholder diversity and changes over time	Section 2.1.1 of the MR and verification interviews confirm that the project continues to engage a diverse range of stakeholders, including women, youth, and vulnerable households/13/10/. The demographic composition of stakeholders remains consistent with that identified during validation. No significant shifts or exclusions were noted during this monitoring period.
Expected changes in well-being	Evidence was drawn from the Monitoring Report and stakeholder interviews during the site visit. Reported co-benefits include reduced indoor air pollution, time savings in fuel collection, and reduced household fuel expenses. These impacts were consistently reported by ICS users and are in line with project expectations. The verification team finds the expected well-being improvements to be credible and supported.
Location of stakeholders	It has been observed and verified during on-site audit that the project activity is in line with the monitoring plan/5/ and there is no change in the location of stakeholders.

Location of resources	The project does not involve access to or use of shared or contested natural resources. The ICS units operate independently within households and do not require external resource extraction. This criterion is not applicable to the project and was confirmed through field observations and MR review.

4.2.2 Stakeholder Consultation and Ongoing Communication

Ongoing consultation	The verification team reviewed MR Sections 2.1.2 and Appendix IV, which document regular stakeholder engagement activities including awareness programs, household registration, staff training, and sampling surveys[/13/]. Ongoing communication mechanisms were confirmed during field interviews, including household visits and follow-ups conducted by the in-country team. The project maintains effective channels for continuous consultation throughout the implementation and monitoring process. Therefore, on the basis of evidence submitted it has been concluded that PP has maintained an effective communication channel.
Date(s) of stakeholder consultation	Multiple rounds of stakeholder engagement were conducted during the monitoring period: (i) staff training from 08–11 July 2024; (ii) household registration from 19 August–31 October 2024; (iii) distribution follow-ups from 23 September–31 October 2024; and (iv) household survey from 15–17 January 2025. All consultations were documented in the MR and verified through field evidence
Communication of monitored results	The project proponent communicated project objectives and survey intent in local languages via trained field teams. CL 05 was raised to assess more detail related to this. Prior to ICS distribution, awareness sessions were conducted in each ward to explain the benefits and usage of ICS units. During the monitoring survey, enumerators clearly communicated the purpose and data collection methods to selected households

Consultation records	Evidence reviewed includes household registration forms, field visit logs, survey questionnaires, and staff training records. Over 19,000 households were registered and consulted. For the monitoring survey, 96 completed interviews were conducted across 120 sampled households. Data was digitally collected and verified by supervisors and a data entry officer. CL 13 was raised and closed successfully.
Stakeholder input	Inputs from stakeholders were recorded during both the registration and survey phases, including household demographics, contact details, ICS distribution and installation data, and end-user feedback. Feedback on training and project implementation was also documented during staff training sessions, with mechanisms in place to incorporate suggestions into future planning. It has been observed that there has been no major grievance received during this verification.

4.2.3 Free, Prior, and Informed Consent

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Consent	The assessment team reviewed Section 2.1.2 of the Monitoring Report and consulted field reports from stakeholder interactions/13/. Consent was obtained through direct engagement with end users during awareness-raising sessions and the household registration process, conducted prior to stove distribution and also from the local government. Consent was voluntary, and participants were provided with information on the project, stove usage, and participation rights in local languages. Field interviews during the site visit confirmed that no coercion or undue influence was applied. The process is deemed consistent with FPIC principles.
Outcome of FPIC discussion	No concerns or objections regarding participation were reported. The verification team concludes that FPIC was appropriately obtained and documented in line with the VCS Program and project safeguards.

4.2.4 Grievance Redress Procedure

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Grievance received and steps taken to resolve the grievance including the outcomes of the resolution	The verification team reviewed Section 2.1.4 of the Monitoring Report. No major grievances/13/ were reported or recorded during the current monitoring period. This was confirmed through interviews with project staff and a review of field documentation. Based on this evidence, it is concluded that the grievance redress process is in place, functional, and no resolution actions were required during this periodIt is crucial to emphasize that the responsibility for addressing grievances related to the grouped project activity is shared between the local implementation partners and the project proponent, as verified based on interviews.
Grievance redress procedure	The project has established a Grievance Redress Mechanism with a Standard Operating Procedure/17/, as described in the Monitoring Report. This includes registration and tracking of complaints by the in-country project manager and resolution with the support of field assistants using traditional and customary approaches. The verification team finds the GRM to be appropriate, accessible, and consistent with VCS Program requirements

4.2.5 Public Comments

Through onsite visit call[/10/], confirmation is obtained from Project Proponent that no public comment received for this grouped project. Through virtual demonstration, it is confirmed that no email communication are received from Verra in relation to this aspect till date. The same claim is confirmed in section 2.1.5 of the monitoring report[/8/].

Comments received	Actions taken by the project proponent	Evidence gathering activities, evidence checked, and assessment conclusion
From the evidence gathered, no public comment has been received	N/A	N/A

4.2.6 Risks to Local Stakeholders and the Environment

4.2.6.1 Management Experience

Evidence gathering activities: The verification team reviewed Section 2.2.1 and 2.6 of the Monitoring Report, including the organizational structure, task allocation table, and team roles outlined for implementation and monitoring.

Evidence checked: The project is managed by TEM PNG Cookstoves Project Pte. Ltd., which has been active in the carbon project space, checked through internet search. The MR details clearly define roles and responsibilities of the Director of International Projects, Country Manager, Field Coordinator, and Field Assistants, demonstrating structured delegation were, also cross verified during the interviews.

Field teams have been recruited locally and trained[/17/] in stove distribution, data collection, quality assurance, and monitoring survey procedures.

Assessment conclusion: The verification team concludes that the project proponent demonstrates sufficient expertise and capacity to manage the project. Where additional expertise is needed (e.g. on socio-economic dynamics), TEM engages with local experts and academic institutions, verified through the PP interview. The management structure and operational procedures are robust and suitable for the scale and nature of the project activity.

4.2.6.2 Risk Assessment

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Natural and human induced risks to stakeholders' wellbeing	The Monitoring Report (Section 2.2.2) indicates that no natural or human-induced risks were identified for stakeholders' wellbeing. The ICS technology improves indoor air quality and safety by reducing smoke and the risk of burns compared to baseline open-fire cooking as verified through the monitoring survey and onsite audit. Site observations and household interviews confirmed improved safety and no identified adverse impacts. The verification team agrees with the assessment and concludes no such risks are present.
Risks to stakeholder participation	The MR acknowledges that voluntary participation could present a risk of low adoption. Mitigation strategies include ongoing awareness campaigns and end-user education on ICS benefits. Field observations confirmed high acceptance levels and no indication of forced participation. The verification team confirms adequate measures are in place to maintain engagement with the help of wariness training and monitoring practices .

Working conditions	The MR outlines that safe working conditions were maintained for field staff involved in awareness and ICS distribution. Risk mitigation included the involvement of community leaders, police, and private security. Based on interviews and reviewed SOPs, the verification team confirms that adequate procedures were in place and working conditions met safety expectations, the safety procedures were also checked during the onsite audit for the project.
Safety of women and girls	The MR identifies no safety risk to women and girls; instead, the project enhances safety by reducing firewood collection trips and replacing unsafe open fires. Field feedback from women beneficiaries corroborated this. The verification team concludes this positive outcome is well-substantiated
Safety of minority and marginalized groups, including children	No risks were identified. ICSs reduce exposure to dangerous open fires, especially for children. The MR and site visits confirmed that the project does not discriminate and is inclusive of vulnerable groups. The verification team concurs with the risk assessment
Pollutants (air, noise, discharges to water, generation and release of hazardous materials and chemical pesticides and fertilizers	The project reduces air pollutants by improving combustion efficiency. ICSs are made of stainless steel, are recyclable, and do not produce hazardous waste. No water or chemical discharges occur. Based on design documents and MR review, the verification team finds the environmental impact is beneficial, with no pollutant risks identified

4.2.7 Respect for Human Rights and Equity

4.2.7.1 Labor and Work

Item	**Evidence gathering activities, evidence checked, and assessment conclusion**
Discrimination	The verification team reviewed the TEM Employee Handbook and Modern Slavery Policy, as cited in the MR. The project prohibits discrimination based on race, gender, religion, age, or disability. Grievance procedures are in place, and field interviews confirmed no reported cases of discriminatory behaviour. The team concludes that risks are recognized and appropriate mitigation exists

Sexual harassment	As per Section 13 of the TEM Employee Handbook, any form of sexual harassment is explicitly forbidden and reportable. Field staff are informed about reporting channels and protections. The verification team confirms the presence of adequate training and enforcement mechanisms. No incidents were reported during the monitoring period
Gender equity in labor and work	The project actively promotes gender equity. Of the 24 local individuals hired, 37.5% are women. Employment and remuneration are based on qualifications and merit. Based on review of team rosters and interviews, the verification team finds that hiring practices are gender-inclusive and fair
Forced labor	The project adheres to the Modern Slavery Policy, which explicitly prohibits forced labor and applies equally to staff and third-party workers. No evidence of forced labor was observed or reported during verification. The team concludes the project complies with international labor standards
Child labor	The use of child labor is strictly prohibited under the project's Modern Slavery Policy. Verification interviews and document review confirm no involvement of minors in project activities. Training records and field staffing lists show adherence to this policy.
Human trafficking	Human trafficking is also explicitly opposed in the project's labor governance framework. The verification team found no indicators or reports of trafficking-related risk and confirms that staff were appropriately sourced and contracted.

4.2.7.2 Human Rights

Risks identified	Evidence gathering activities, evidence checked, and assessment conclusion

No risks to human rights were identified during this monitoring period.	The verification team reviewed Section 2.3.3 of the Monitoring Report, along with field team policies, community engagement procedures, and the stakeholder consultation process. The project operates in rural Papua New Guinea, where Indigenous and local communities are the primary beneficiaries. All participation was voluntary, and no displacement or interference with customary rights was reported. The verification team confirmed through household interviews and consultation records that the project respected the rights of Indigenous Peoples and local communities. The project is in alignment with relevant human rights standards, including the UN Declaration on the Rights of Indigenous Peoples and ILO Convention 169. The verification team concludes that the project poses no risk to human rights and is compliant with VCS Program safeguards.

4.2.7.3 Indigenous Peoples and Cultural Heritage

Risks identified	Evidence gathering activities, evidence checked, and assessment conclusion
No risks related to Indigenous Peoples or cultural heritage were identified.	The verification team reviewed Section 2.3.3 of the Monitoring Report, which confirms that the project is implemented in areas predominantly inhabited by Indigenous Peoples, and that no displacement, land-use change, or interference with cultural practices has occurred. Stakeholder consultations were carried out in local languages with support from community leaders, and the project activities—distribution and use of ICS units—do not impact sacred sites, traditional land, or cultural heritage. The project was welcomed by local communities and aligns with customary norms. The verification team confirms that no adverse impacts on Indigenous rights or cultural heritage were identified and that the project complies with the VCS safeguard requirements for Indigenous Peoples and cultural heritage.

4.2.7.4 Property Rights

Risks identified	Evidence gathering activities, evidence checked, and assessment conclusion

No risks related to property rights were identified.	The verification team reviewed Section 2.3.4 of the Monitoring Report and verified that the project does not involve land acquisition, land use changes, or the use of shared community resources. The distribution and use of improved cookstoves take place entirely within individual households and do not involve collective or customary land or territorial access. There is no restriction on resource access, and ICS distribution was conducted on a voluntary basis. During field visits, no disputes or grievances related to property rights were raised. The verification team concludes that the project does not pose any risk to legal or customary property rights and complies with VCS safeguard requirements.

4.2.7.5 Benefit Sharing

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Summary of the benefit sharing plan	The verification team reviewed Section 2.3.5 of the MR and confirmed that the benefit sharing plan involves equitable access to ICS distribution for all eligible households within the project boundary. Households are required to complete both the Household Registration and ICS Distribution processes. Upon successful onboarding, each household receives an ICS free of charge. These cookstoves are intended to improve indoor air quality, reduce time and costs associated with fuel collection/purchase, and mitigate forest degradation
Benefit sharing during the monitoring period	During the period from 23 September 2024 to 31 October 2024, a total of 19,206 ICS units were distributed to unique households. Verified benefits observed included: (i) reduced time and money spent on firewood, (ii) improved indoor air quality due to higher combustion efficiency, and (iii) reduced degradation of surrounding forest resources. These benefits were confirmed through review of household survey responses, distribution records, and field observations

4.2.8 Ecosystem Health

Item	Evidence gathering activities, evidence checked, and assessment conclusion

Impacts on biodiversity and ecosystems	The verification team assessed Section 2.4 of the MR. The ICS technology reduces the need for firewood collection and combustion, leading to reduced pressure on local forests. As the project does not involve land-use change, infrastructure development, or introduction of invasive species, no adverse impacts on biodiversity and ecosystems were identified.
Soil degradation and soil erosion	The ICS technology is deployed at household level without disturbing soil or vegetation. No construction or large-scale ground disturbance was observed. Thus, the project has no negative impact on soil degradation or erosion.
Water consumption and stress	The ICSs do not consume or discharge water as part of their operation. There is no infrastructure in place requiring water usage. Therefore, no additional water consumption or stress was observed during the monitoring period

4.2.8.1 Rare, Threatened, and Endangered species

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Species or habitat	N/A. The nature of the project implementation is through the dissemination of improved cookstove to replace the traditional baseline stove which does not present risk for local species or habitat
Areas needed for habitat connectivity	NA

4.2.8.2 Introduction of Species

Species introduced	Evidence gathering activities, evidence checked, and assessment conclusion
NA	

Existing invasive species	lence gathering activities, evidence checked, and assessment conclusion
NA	

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Invasive species	**Evidence gathering activities, evidence checked, and assessment conclusion**
NA	

4.2.8.3 Ecosystem conversion

Item	Evidence gathering activities and evidence checked
Ecosystem conversion	N/A. The nature of the project implementation is through the dissemination of improved cookstove to replace the traditional baseline stove which does not present as risk to local ecosystem.

4.3 Accuracy of Reduction and Removal Calculations

The monitoring has been carried out in accordance with the provision of monitoring plan; the assessment team reviewed if:

- The monitoring of reductions in GHG emissions resulting from the VCS project activity is verified against the monitoring plan contained in the registered and revised VCS-PD[/5//9/] and the applied methodology[/8/].
- All parameters stated in the monitoring plan, the applied methodologies and relevant standards and requirements have been sufficiently monitored and updated.
- The responsibilities and authorities for monitoring and reporting were in accordance with the responsibilities and authorities stated in the monitoring plan.

The monitoring system and all applied procedures are in compliance with the monitoring plan contained in the revised VCS-PD[/8/] and the applied methodology VM0050 V1.0[/8/], based on the information included in the final monitoring report, there are several procedures for data collecting depending on the methodology applicable for each step of the project.

PP Sampling procedure: The verification team assessed the sampling plan and implementation as carried out by the project proponent during the current monitoring period. The sampling was conducted in accordance with the CDM Sampling Standard and guidelines and the requirements of methodology VM0050 v1.0.
The sampling aimed to measure key parameters such as:
- nj,k,y: Proportion of ICSs in use during the monitoring period,
- BCp and BCb: Annual fuelwood consumption under project and baseline scenarios,
- $\eta_{new,avg,y}$: Thermal efficiency of the ICS, verified from third-party lab data,
- User behaviour and stove stacking factors.
A probability-based random sampling approach was applied, with sampling conducted at the Grouped Project level. The project used simple random sampling from the complete ICS distribution database using unique identifiers, ensuring statistical representativeness and eliminating selection bias. The information was also cross verified from the onsite visit.

Baseline and Project Sampling Execution:
The sampling campaign to determine fuel consumption was conducted in October 2022 by an independent firm, Simi Oa Consultancy Services, to eliminate operator bias, as verified from registered PD and validation report. It followed the Kitchen Performance Test protocol.
Baseline (OFC) group:

- Sample Size: 71 households
- Relative precision: 12.36%
- Action taken: Applied lower bound estimate of 5.24 tonnes/year of wood fuel

- Project (ICS) group:
  - Sample Size: 60 households
  - Relative precision: 12.98%
  - Action taken: Applied upper bound estimate of 1.54 tonnes/year

These conservative bounds were applied in line with the CDM sampling guidelines when the 90/10 confidence/precision target was not met.

Stratification and Homogeneity Consideration: Stratification was not applied, as only a single ICS model was deployed during the monitoring period. However, the proponent included a clause to stratify samples in future monitoring periods if multiple ICS types are used.

The grouped project sampling design accounts for technological and geographic homogeneity, complying with the eligibility and applicability requirements for grouped sampling under methodology and CDM standards.

Usage Parameter Sampling (nj,k,y): For stove usage (nj,k,y), a separate random sample of 120 households was drawn. Of these, 96 interviews were successfully completed, with 24 non-responses documented due to household unavailability or non-cooperation. The response rate of 80% was deemed acceptable under CDM standards, and reasons for attrition were recorded transparently. Survey teams used digital tools (Survey CTO) and GPS-verified records to confirm correct household identity, stove presence, and operational status. Additional layers of QA included back-checks and manual verification of photos and responses by the in-country technical team.

Data Quality Assurance (QA/QC)

- Enumerators underwent structured training and refresher sessions.
- Field protocols were enforced to reduce interviewer bias.
- Survey entries were cross-checked by the supervisor and technical teams.
- Redundant and incorrect data points were flagged and removed before analysis.

The verification team reviewed the raw dataset, sampling logic, calculation sheets, and supporting photos during verification, a remote call was conducted with PP to verify the Powersolve platform (Used by TEM Senior Team to analyze data and monitor the project activity at a large scale. The PowerSolve Platform pulls data from SurveyCTO and matches data from different surveys to ensure no duplicates are recorded in the final count of ICS Distributed.)

Assessment Conclusion: The sampling methodology, execution, and documentation were found to be Methodologically sound, Statistically valid, Conservatively applied, and In full compliance with CDM and VCS guidance.

CL13 was raised related to sampling details and was closed successfully.

The use of conservative bounds where statistical precision was not achieved adds robustness to the emission reduction estimates. The data collection process and quality control measures were transparently documented and verified through on-site review and interviews.

Emission reduction (ER)

The Group project activity calculates emission reductions inline with the applied methodology VM0050,v1.0

Baseline emissions are calculated using equation 1:

$BEy = ECi,y \times Nj,k,y \times nj,k,y \times (EFb,i,CO_2 \times fNRB,y + EFb,i,non\text{-}CO_2)$

Where:

ECi,y is the annual energy consumption per device (in TJ),

Nj,k,y is the number of devices distributed per batch,

nj,k,y is the proportion of devices in use,

EFb,i,CO_2 = 112 tCO_2/TJ and EFb,i,non-CO_2 = 9.46 tCO_2e/TJ (IPCC 2019), fNRB (fraction of non-renewable biomass) = 60.12%, conservatively adjusted from an original estimate of 81.24% based on Tool 30 v4

The average household wood fuel consumption in the baseline scenario (OFC) was determined through a KPT measurement campaign involving 71 samples. Due to a 12.36% precision, the lower bound value of 5.24 tonnes/HH/year was applied.
The energy consumption ECi,y was derived using:
Equation 2:
ECi,y = BCb,i,y × NCVb,i
= 5.24 t × 0.02 TJ/t = **0.08 TJ/HH/year**
Project emissions:
Project emissions reflect the emissions from improved cookstove (ICS) use. These are calculated using:

Equation 7:
PEenergy,y = BCp,j,k,y × Nj,k,y × NCVp,j × nj,k,y × (EFb,i,CO_2 × fNRB,y + EFb,i,non-CO_2)

Where:

BCp = 1.64 (batch 1) and 1.54 (batch 2) tonnes/year per household
NCV = 0.0156 TJ/tonne,
Same emission factors and fNRB as baseline.

The values were aggregated for two batches:

Batch 1: 4,647 ICSs → PE = 1,909 tCO_2e

Batch 2: 14,559 ICSs → PE = 4,090 tCO_2e

Total Project Emissions: 5,999 tCO_2e

Leakage:
As per VM0050, leakage due to fossil fuel displacement was addressed using a default deduction factor:
LEy = (BEy – PEy) × 0.05
i.e., 5% deduction applied, equivalent to 708 tCO_2e

Emission Reductions (ERy)

Net GHG emission reductions are calculated using:
Equation 11:

ERy = (BEy − PEy) × 0.95 − LERB,y

Where LERB,y = 0 (no renewable biomass leakage)

Batch	BEy (tCO_2e)	PEy (tCO_2e)	LE	ERy (tCO_2e)
1	6,226	1,909	216	4,101
2	13,924	4,090	492	9,342
Total	20,150	5999	708	13,443

The data and parameters fixed (ex-ante):

S.No	Parameter	Value	Means of verification
1	$EF_{b,i,CO2}$ CO2 emission factor for fuel used by baseline device type i in the baseline scenario.	112 tCO_2/TJ	Default value from 2019 IPCC Guidelines, Stationary Combustion, Chapter 2, inline with methodology option 3, therefore acceptable.
2	EFb,i,non-CO_2 (CH_4 and N_2O expressed as CO_2e)	9.46 tCO_2e/TJ	Same as above. Default non-CO_2 emissions for wood combustion. inline with methodology option 3, therefore acceptable.
3	NCVb,i / NCVp,j (Net Calorific Value of wood fuel)	0.0156 TJ/tonne	Default from 2019 IPCC Guidelines based on air-dried wood fuel. Applied consistently for both baseline and project fuel use.
4	BCex-ante,b,i (Baseline fuel use per household)	5.24 tonnes/year	Determined through KPT by third-party (Simi Oa Consultancy Services), with lower bound applied for conservativeness (precision = 12.36%), inline with methodology requirement option 1, Assessment team has assed the KPT survey and it is found to be appropriate.

5	ηold,avg (Efficiency of baseline stoves)	0.15	Methodology default for three-stone fire (CDM TOOL33 v2). No efficiency tests conducted due to the known technology type and absence of improved combustion or flue gas ventilation
6	Technology lifetime	10 years	The value used has been verified from the manufacturing specifications of the improved cookstove
7	**fNRB** (Fraction of non-renewable biomass)	60.12%	Derived using CDM Tool 30 v4. Original fNRB calculated as 81.24%, discounted by 26% in this monitoring period.
8	Household size (Hhi)	4.85 equivalent male adults	Based on a 2022 field study audited by PNG University of Technology. Used to normalize fuel consumption per household in the baseline and project scenarios

The assessment team confirms that:
- All fixed parameters are derived from credible sources, either default values in the methodology/IPCC guidelines or conservatively calculated based on verifiable national or project-level studies.
- Their use was consistent with the applied methodology and sampling standard (CDM v9.0).
- Where statistical uncertainty existed, conservative bounds were applied to ensure the environmental integrity of the reported emission reductions.

The data and parameters monitored (ex-post):

1. Parameter: Nj,1,y : Number of commissioned project devices of type j from batch 2 in year 1
Nj,2,y: Number of commissioned project devices of type j from batch 2 in year 1

Means of verification	Measuring /Reading /Recording frequency	The monitoring frequency is annual, whenever a new ICS will be distributed.
	Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes. The frequency applied aligns with the plan stated in the PDD and methodology.
	Monitoring equipment	Not applicable. Value is determined through administrative records (Distribution ledger and distribution logs).
	Is accuracy of the monitoring equipment as stated in the monitoring plan? If the monitoring plan does not specify the accuracy of the monitoring equipment, does the accuracy of the monitoring equipment comply with local/national standards, or as per the manufacturer's specification?	Not applicable – no monitoring equipment involved. Accuracy relies on documentation.
	Calibration frequency /interval:	Not applicable.
	Is the calibration interval in line with the monitoring plan and/or methodology? If the monitoring plan does not specify the frequency of calibration, is the selected frequency in accordance with the local/national standards, or as per the manufacturer's specifications?	Not applicable.

	Is the calibration of measuring equipment carried out by an accredited person or institution?	Not applicable.
	Is(are) calibration(s) valid for the whole reporting period?	Not applicable.
	How were the values in the monitoring report verified?	Value of batch 1 is 4,647 units and batch 2 14,559 was verified by reviewing distribution records and purchase invoices for respective batches. Cross-checked with the cumulative stove distribution list maintained by the project proponent and confirmed during field audit.
	If applicable, has the reported data been cross checked with other available data?	Yes. Verified against distribution database, invoice records, and field visit observations.
	Does the data management ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Yes. QA/QC procedures include maintaining electronic and physical distribution records, verified by the in-country field coordinator and checked by the verification team. Consistent with MR QA/QC description
	In case project participants have temporarily not monitored the parameter, has either deviation?	No deviation. The parameter was monitored and reported in accordance with the plan.
Findings	No finding was raised.	
Conclusion	The verification team concludes that the parameter $\mathbf{N_{j,k,y}}$ has been correctly monitored and reported. The data source is valid, the reporting frequency aligns with the methodology and PDD, and adequate QA/QC measures are in place. The value is accepted for use in emission reduction calculations without any corrective action.	

2. Parameter:
Date of commissioning of batch 1
Date of commissioning of batch 2

Means of verification	Measuring /Reading /Recording frequency	. Fixed; recorded once at the time of commissioning/distribution of the last ICS in batches
	Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes. The date was determined based on the internal log maintained by the project proponent, as required under the monitoring plan and consistent with VM0050, which allows grouping of devices by commissioning date.
	Monitoring equipment	Not applicable – the value is derived from administrative distribution logs.
	Is accuracy of the monitoring equipment as stated in the monitoring plan? If the monitoring plan does not specify the accuracy of the monitoring equipment, does the accuracy of the monitoring equipment comply with local/national standards, or as per the manufacturer's specification?	Not applicable.
	Calibration frequency /interval:	Not applicable.

	Is the calibration interval in line with the monitoring plan and/or methodology? If the monitoring plan does not specify the frequency of calibration, is the selected frequency in accordance with the local/national standards, or as per the manufacturer's specifications?	.
	Is the calibration of measuring equipment carried out by an accredited person or institution?	Not applicable.
	Is(are) calibration(s) valid for the whole reporting period?	Not applicable.
	How were the values in the monitoring report verified?	Batch 1 – 30/09/2024 Batch 2 – 31/10/2024 It was verified against the internal ICS distribution records Both these dates are the dates on the which the last stoves under the batches were distributed, leading to complete distribution.
	If applicable, has the reported data been cross checked with other available data?	Yes. Cross-checked with the cumulative distribution sheet, survey records, and household installation logs. The date was confirmed during document review and field verification.
	Does the data management ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Yes. Data transfer was done digitally with backup logs; internal tracking systems are in place and reviewed by the field coordinator.

	In case project participants have temporarily not monitored the parameter, has either deviation?	No deviation. The parameter was monitored and reported according to the PDD and MR.
Findings	No finding raised.	
Conclusion	The parameter is accurately determined and reported. It is compliant with the methodology's requirements for batch grouping and device commissioning, and no discrepancies or deviations were noted during verification.	

3. Parameter

nj,1,y - Proportion of commissioned project devices of type j from batch 1 that are still being used regularly in year y

nj,2,y - Proportion of commissioned project devices of type j from batch 2 that are still being used regularly in year y

Means of verification	Measuring /Reading /Recording frequency	Annually. As per the monitoring plan, this parameter is monitored each year during the crediting period using surveys
	Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes. Frequency is consistent with the guidance under VM0050 and CDM Sampling Standard v9.0. The sampling was carried out once during the monitoring period (January 2025), as expected.

	Monitoring equipment	Not applicable – the parameter is estimated through household surveys. Survey CTO were used for field data collection. Along with visual inspection of the kitchen observations and interview the primary cook. The similar practice was also followed by the assessment team when from the VVB site, All the end users interviewed were inspected for there kitchen observation to confirm the usage of the project stove and photographic evidence were collected with the end users. All the staff members who conducted the interview were also trained by the PP, the staff training records have also been cross checked meeting the methodology requirement.
	Is accuracy of the monitoring equipment as stated in the monitoring plan? If the monitoring plan does not specify the accuracy of the monitoring equipment, does the accuracy of the monitoring equipment comply with local/national standards, or as per the manufacturer's specification?	Not applicable. Accuracy is ensured through digital survey methods, field team training, and quality checks.
	Value applied	Batch 1 – 0.84 Batch 2 – 0.9

			During the local stakeholder Consultations and household surveys, the project assessed cooking practices and preferences to ensure that the distributed ICSs fully meet the cooking needs of the target population. The ICS units were selected based on these assessments and stakeholder feedback to replace open-fire cooking effectively for the primary cooking needs. Implementation of Support Activities: The project proponent provided instructional flyers, in-person training during distribution, and demonstrations on operating and maintaining the ICS units for cooking local foods. Troubleshooting guides and maintenance tips were shared in Tok Pisin and English, the commonly used languages in the project area. During twice-yearly visits, the PP's in-country team conducted inspections, minor repairs, and user re-trainings to ensure continued effective use of the ICS units. Provision of a Communication Channel: Multiple channels were made available for stakeholders and end-users to contact the project proponent for support: Dedicated project website: https://www.tasmanenvironmental.com. au/pngcookstoves/

			Project email: pngcookstoves@tem.com.au Country Manager email: michael.kisombo@tem.com.au PNG Office phone number (toll-free within PNG): +675 71532292 These channels allow users to request maintenance, report issues, and seek operational guidance throughout the project's lifetime. Evidence: Stakeholder consultation records, Flyers and materials provided to end-users, Distribution and maintenance visit records, Communication channel details and evidence of use
		Calibration frequency /interval:	Not applicable
		Is the calibration interval in line with the monitoring plan and/or methodology? If the monitoring plan does not specify the frequency of calibration, is the selected frequency in accordance with the local/national standards, or as per the manufacturer's specifications?	

	Is the calibration of measuring equipment carried out by an accredited person or institution?	Not applicable
	Is(are) calibration(s) valid for the whole reporting period?	Not applicable
	How were the values in the monitoring report verified?	The monitoring team conducted household surveys from 15–17 January 2025. Out of 120 sampled ICS households, 96 interviews were completed. Based on the survey, the usage rate ($n_{j,k,y}$) was determined as 1, as all verified households had their ICS installed and in regular use. Values were verified through direct observation, stove condition, and user confirmation.
	If applicable, has the reported data been cross checked with other available data?	Yes. Cross-checked with the distribution database, installation records, and GPS-tagged household visit logs. Field photos and enumerator notes confirmed ICS presence.
	Does the data management ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Yes. QA/QC measures include supervisor checks, validation of survey entries, and real-time data transfer using digital tools. Results were reviewed and approved by the in-country technical lead.

	In case project participants have temporarily not monitored the parameter, has either deviation?	No deviation. Monitoring was completed in line with the plan, and sample attrition (24 households) was documented with valid reasons.
Findings	No finds raised.	
Conclusion	The verification team confirms that the parameter $n_{j,k,y}$ was appropriately monitored, sampled, and conservatively applied. The value used in emission reduction calculations is justified and compliant with VCS and CDM standards.	

4. Parameter:
BCb,i,y: Fuel used per baseline device type i during year y

Means of verification	Measuring /Reading /Recording frequency	Every five years, as per the monitoring plan and VM0050 methodology.
	Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes. The frequency is compliant with VM0050, which allows baseline fuel consumption to be reassessed every five years using a representative sample and CDM Sampling Standard.
	Monitoring equipment	Not applicable – data collected through field measurement surveys using Kitchen Performance Test (KPT) protocol. Fuel was weighed using calibrated spring scales during the controlled cooking test campaign.

	Is accuracy of the monitoring equipment as stated in the monitoring plan? If the monitoring plan does not specify the accuracy of the monitoring equipment, does the accuracy of the monitoring equipment comply with local/national standards, or as per the manufacturer's specification?	Yes. While the exact calibration specifications are not detailed, the use of standardized KPT procedures and professional field teams (third-party consultant Simi Oa) meets accepted good practice
	Calibration frequency /interval:	Not explicitly stated, but KPT scale calibration is industry standard before and after each campaign.
	Is the calibration interval in line with the monitoring plan and/or methodology? If the monitoring plan does not specify the frequency of calibration, is the selected frequency in accordance with the local/national standards, or as per the manufacturer's specifications?	Yes, implicitly. The methodology requires robust field procedures; the verification team found no evidence of deviation or equipment inaccuracy.
	Is the calibration of measuring equipment carried out by an accredited person or institution?	Equipment was operated by trained enumerators under supervision of a third-party consultancy. While formal accreditation is not specified, the team was qualified and independent.
	Is(are) calibration(s) valid for the whole reporting period?	Yes. The measurement campaign was conducted in October 2022 and is valid for the current and upcoming monitoring periods until new data are collected

		How were the values in the monitoring report verified?	The campaign involved 71 valid samples for open fire cooking (OFC) across two provinces. A precision of 12.36% was achieved. In line with CDM Sampling Standard v9.0, the lower bound value of 5.24 tonnes/year per household was applied. The team reviewed raw data logs, sampling plans, and consultant report (TEM 2025_Cookstoves Highlands region report_v3.pdf).
		If applicable, has the reported data been cross checked with other available data?	Yes. The value was cross-validated with the summary of campaign data, the baseline household registration list, and the field report. Sample consistency and location diversity were confirmed.
		Does the data management ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Yes. Data entry and reporting follow the QA/QC plan outlined in the MR. Values were double-checked, and conservative bounds applied to ensure robustness.
		In case project participants have temporarily not monitored the parameter, has either deviation?	No deviation. The parameter was monitored before the start of crediting and is valid for up to five years, as per the methodology.
Findings	CL 08 was raised and closed successfully.		
Conclusion	The parameter $BC_{ex-ante,\beta,i}$ was appropriately derived, verified using supporting data, and applied conservatively in accordance with methodology VM0050 and CDM Sampling Standard v9.0. The verification team accepts the use of 5.24 t/year as valid and robust.		

4. ηnew,avg,y: Weighted average efficiency of project devices in year y

Means of verification	Measuring /Reading /Recording frequency	Recorded at time of commissioning and annually thereafter. However, for the current monitoring period, no efficiency degradation monitoring was required since the ICSs were recently distributed (less than one year).
	Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes. The frequency and approach are consistent with the requirements of VM0050 v1.0 and AMS-II.G v13.1. The methodology permits using the manufacturer-declared efficiency when ICSs are newly distributed and in use for less than one year.
	Monitoring equipment	Standardized laboratory-based Water Boiling Test (WBT) methodology, conducted by the manufacturer and supported by ISO-compliant QA systems.
	Is accuracy of the monitoring equipment as stated in the monitoring plan? If the monitoring plan does not specify the accuracy of the monitoring equipment, does the accuracy of the monitoring equipment comply with local/national standards, or as per the manufacturer's specification?	Yes. The value was derived from manufacturer-certified WBT testing. No degradation testing was required, as this is the initial monitoring period.

	Calibration frequency /interval:	Not applicable for this period, as efficiency testing is not repeated during the first year post-distribution.
	Is the calibration interval in line with the monitoring plan and/or methodology? If the monitoring plan does not specify the frequency of calibration, is the selected frequency in accordance with the local/national standards, or as per the manufacturer's specifications?	Yes, implicitly. The project will conduct annual testing in future years as required by AMS-II.G, with Standard WBT applied to samples from each batch.
	Is the calibration of measuring equipment carried out by an accredited person or institution?	Yes. The efficiency value of 0.52 is based on WBTs performed by the manufacturer, using certified equipment under recognized quality control (e.g. ISO certification of process). In line with the clarification mentioned with methodology, the manufacture specification for WBT was also supported by accredited independent laboratory results. CL 14 was raised and closed successfully.
	Is(are) calibration(s) valid for the whole reporting period?	Yes. The ICSs were all installed during the monitoring period, hence the manufacturer-declared efficiency is valid for the entire duration.

	How were the values in the monitoring report verified?	The efficiency value was checked against the manufacturer test report and validated as compliant with the WBT protocol and the efficiency thresholds defined in the methodology. Field cross-checks confirmed the stoves distributed match the tested model.
	If applicable, has the reported data been cross checked with other available data?	Yes. Cross-verified against stove specifications submitted in the project documentation and supported by third-party verified lab test reports included in the project annexes.
	Does the data management ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Yes. The project uses digital tracking and batch-wise assignment of efficiency values to ICS distributions, and documentation is maintained in line with monitoring requirements.
	In case project participants have temporarily not monitored the parameter, has either deviation?	No deviation. The methodology permits the use of the manufacturer value for the first year after ICS commissioning. There is no requirement for degradation measurement during this initial period.
Findings	CL 14 was raised and closed successfully.	
Conclusion	The efficiency parameter $\eta_{new,avg,y}$ = 0.52 was applied correctly and in line with methodological provisions for new ICS deployments. The verification team finds no deficiencies in the parameter estimation or usage and confirms it is valid for this monitoring period.	

5. Parameter:

BCp,j,1,y - Average quantity of fuel used by project device type j from batch 1 during year y

BCp,j,2,y - Average quantity of fuel used by project device type j from batch 2 during year y

Means of verification	Measuring /Reading /Recording frequency	The parameter is measured biannually or annually, in accordance with Option 1 of the methodology. For this monitoring period, the parameter was derived from Kitchen Performance Tests conducted in November 2022 and July 2025. The November 2022 KPT results are valid until November 2024, as per Section 9.2 of VM0050 v1.0. To ensure continued compliance beyond this period, the project proponent conducted an updated KPT campaign in July 2025. Out of the two BCp values obtained, the PP has decided to apply the highest BCp value for each the respective batches, that will lead to the most conservative emission reduction estimations (ERs) throughout the monitoring period.
	Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes. The frequency and approach are consistent with VM0050 , which require representative sampling and periodic reassessment.

	Monitoring equipment	No physical monitoring device used directly for this parameter. Fuel weights were recorded during a Kitchen Performance Test (KPT) following standard procedures.
	Values applied	Batch 1 = 1.61 Batch 2 = 1.54
	Is accuracy of the monitoring equipment as stated in the monitoring plan? If the monitoring plan does not specify the accuracy of the monitoring equipment, does the accuracy of the monitoring equipment comply with local/national standards, or as per the manufacturer's specification?	Yes. While not explicitly mentioned, field teams used calibrated spring scales for fuel weighing. Results are considered reliable.
	Calibration frequency /interval:	Not applicable for this parameter, but implied to follow standard KPT procedures.
	Is the calibration interval in line with the monitoring plan and/or methodology? If the monitoring plan does not specify the frequency of calibration, is the selected frequency in accordance with the local/national standards, or as per the manufacturer's specifications?	Yes, implicitly. The methodology and MR accept equipment operated by third-party enumerators under proper QA protocols

	Is the calibration of measuring equipment carried out by an accredited person or institution?	Measurements were conducted by trained staff from a third-party consultant, Simi Oa Consultancy Services. Accreditation is not mandatory, but professional expertise was demonstrated.
	Is(are) calibration(s) valid for the whole reporting period?	No, KPT test needs to be repeated on the biannual basis therefore, It will be done as per the tests.

	How were the values in the monitoring report verified?	The values reported in the monitoring report were verified through the following means: Review of KPT Documentation: The VVB reviewed the KPT reports from both the November 2022 and July 2025 campaigns. This included sample size, methodology applied, statistical precision achieved, and the application of conservativeness deductions where applicable. Photographic and Field Evidence: Photographic evidence from both KPT campaigns was examined to confirm that the same stove model and operational conditions were used, in line with the requirements of VM0050 v1.0. Verification of Conservative Value Application: The project proponent applied the more conservative value between the two KPTs for each batch, which was verified by the VVB as a conservative and compliant approach.
	If applicable, has the reported data been cross checked with other available data?	Yes. Verified against the consultant's final field report, individual household records, photographic evidence and GPS-tagged survey data. No major inconsistencies were found.

		Does the data management ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Yes. QA/QC procedures included supervised data entry, data cleaning, and independent verification of anomalies before final value selection.
		In case project participants have temporarily not monitored the parameter, has either deviation?	The value monitored during the November 2022 KPT was valid until November 2024, in accordance with Section 9.2 of VM0050 v1.0. However, since this left a gap for December 2024, the project proponent conducted a new KPT in July 2025. To address this short gap, the project proponent applied a conservative deviation by using the higher (more conservative) value between the two KPTs for each batch throughout the entire monitoring period. This approach ensures conservativeness in emission reduction calculations and maintains compliance with methodological requirements despite the brief lapse in monitoring coverage.
Findings	CL08 was raised and closed successfully.		
Conclusion	The reported value is valid, conservative, and derived following the required methodological protocols. The verification team accepts this value for emission reduction calculations without reservations.		

Full assessment of value and data used in ER calculation is verified as detailed above. Clarifications and inconsistencies raised, as CLs and CARs have been resolved as provided in each specific parameter.

The assessment team has thoroughly verified the calculation done for the quantification of the emissions reduction for the project activity, under this monitoring period from 23/09/2024 to 31/12/2024.

The assessment team has checked and confirmed the calculations in the calculation sheet and found them to be accurate. The monitoring report is supported by an emission reduction spreadsheet. The consistency and formula were verified and found to be accurate.

4.4 Quality of Evidence to Determine Reductions and Removals

The verification team has assessed the evidence sources, data quality, and procedural controls used in calculating emission reductions during the monitoring period 23 September 2024 to 31 December 2024. The following aspects were examined in line with the guidance:

1. Reliability of Evidence

The data sources used for estimating GHG emission reductions are reliable, traceable, and appropriately referenced:

Fuel consumption data (BCb and BCp) were obtained through a KPT-based measurement campaign conducted by an independent third-party (Simi Oa Consultancy) in October 2022. Sampling plans and raw data are documented and available for review.

Efficiency (ηnew) is based on manufacturer-certified lab tests using the WBT protocol, in line with AMS-II.G and VM0050. Since all ICSs were recently installed, no degradation adjustment was necessary.

ICS usage and operational status (nj,k,y) were verified through household surveys using GPS-referenced data collection via Survey CTO. Cross-verification with distribution records showed alignment.

All evidence reviewed was documented and externally verifiable, with proper archiving and field records maintained.

2. Information Flow from Data Generation to Monitoring Report:

The data flow process from collection to reporting was clearly documented and structured. The steps followed include:

Field data were collected digitally using trained enumerators and mobile data collection tools, reducing risk of manual entry errors.

The survey datasets were transferred to a centralized database and checked by the field coordinator and M&E manager before aggregation.

Emission reduction calculations were performed in a pre-validated Excel-based tool applying equations from AMS-II.G and VM0050.

QA/QC steps were applied for parameter conversions, unit consistency, and formula integrity. The final results are correctly transposed in the MR.

The verification team found no discrepancies between source documentation and reported values in the MR.

Based on the review of primary data, data flow procedures, and QA/QC practices, the verification team concludes that the:
Quantity of evidence is sufficient, and Quality and traceability of evidence is appropriate for the purpose of determining the reported emission reductions.

The data was collected, transferred, and reported in a manner that ensures accuracy, transparency, and conservativeness in line with VM0050 v1.0, Thus a reasonable level of assurance is achieved..

4.5 Non-Permanence Risk Analysis

VCS standard[/1/] section 2.4, non-permanence risk analysis is applicable for Agriculture, Forestry, and Other Land Use (AFOLU) and Geologic Carbon Storage (GCS) projects. Therefore, this section is not applicable for the project activity.

5 VERIFICATION OPINION

5.1 Verification Summary

The project activity VCS – 4416, “Improved Cookstove Grouped Project in Papua New Guinea” has undergone 1st verification with monitoring period from 23/09/2024 to 31/12/2024 (including both days). TEM are responsible for the collection, analysis, aggregation, preparation (conversion factors, assumptions, methodology, calculations) and presentation of activities and related GHG data in its public disclosures (including but not limited to monitoring report, emission reduction calculation sheet, submitted documents/ emails/ communications). Our responsibility of performing this work is to the management of TEM and in accordance with terms of reference agreed with the TEM in the master service agreement. The verification engagement assumes that the GHG data provided to us is complete, sufficient, true and free from material misstatements. SustainCERT disclaims any liability or co-responsibility for any decision a person or an entity would make based on this verification statement towards the issuance and utilization of VCUs hereby verified and certified. The verification was carried out during February 2025 – April 2025 by a team of qualified GHG auditors.

SustainCERT applies its own quality management system and compliance policies for quality control, in accordance with ISO/IEC 17029:2019 – Conformity Assessment Requirements for Validation and Verification bodies providing environmental information (ISO 14065:2020) and greenhouse gas audit (ISO 14064-3:2019) and certifications, and accordingly maintains a comprehensive system of quality control including documented policies and procedures regarding compliance with ethical requirements, professional standards and applicable legal & regulatory requirements.

We have complied with the VCS standard[/1/] and other requirements mentioned, during the verification engagement and maintained independence. SustainCERT was not involved in the preparation of any statements or reports or data except for this Verification Statement and Report. SustainCERT maintains complete impartiality toward stakeholders interviewed during the verification process. SustainCERT did not provide any services to TEM Group and its subsidiaries in the scope of verification that could compromise the independence or impartiality of our work.

5.2 Verification Conclusion

The Verification of the project titled VCS – 4416, "Improved Cookstove Grouped Project in Papua New Guinea", have been undertaken by SustainCERT, as requested by TEM, project proponent for the monitoring period 23/09/2024 to 31/12/2024 (including both the days) as reported in monitoring report[/8/.] The project activity is implemented in accordance with the registered monitored plan in PD[/5/] and the emission reduction from the project activity from the project activity. It is our responsibility to express an independent verification statement on the reported GHG emission reductions from the project activity.

SustainCERT Verification approach was based on understanding the risk associated with reporting of GHG emission data and the controls in place to mitigate these. SustainCERT planned and performed the verification by obtaining evidence and other information and explanations that SustainCERT considered necessary to give reasonable assurance that reported GHG emission reductions are fairly stated.

In our opinion, the GHG emissions reductions reported for the project activity for the period 23/09/2024 to 31/12/2024 are fairly stated in the project monitoring report[/8/]. The GHG emission reductions were calculated correctly based on the approved baseline and monitoring methodology VM0050 Energy Efficiency and Fuel-Switch Measures in Cookstoves, V1.0 and the monitoring plan contained in the PD[/5/].

Verification period: 23/09/2024 to 31/12/2024

Verified GHG emission reductions and carbon dioxide removals in the above verification period:

Vintage period	Baseline emissions (tCO_2e)	Project emissions (tCO_2e)	Leakage emissions (tCO_2e)	Reduction VCUs (tCO_2e)	Removal VCUs (tCO_2e)	Total VCUs (tCO_2e)
23/09/2024 to 31/12/2024	20,150	5,999	712	13,443	-	13,443
Total	20,150	5,999	712	13,443	-	13,443

5.3 Ex-ante vs Ex-post ERR Comparison

Vintage period	Ex-ante estimated reductions/ removals	Achieved reductions/ removals	Percent difference	Explanation for the difference
23/09/2024 to 31/12/2024	103,526	13,443	87%	As per the PD, PP assumed the distribution of 100,000 cookstove, however, the actual distributed stove are 19,206. Therefore, the achieved emission reductions are lower.
Total	103,526	13,443	87%	

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

Use the table below to describe the commercially sensitive information included in the monitoring report to be excluded in the public version.

Section	*Information*	*Justification*	*Assessment method and conclusion*
	N/A		

APPENDIX 2: TABLE OF REFERENCE

S. No	Document Title	Version	Date
1	VCS standard https://verra.org/documents/vcs-standard-v4-7/	4.7	16/04/2024
2	VCS guidelines https://verra.org/wp-content/uploads/2023/08/VCS-Program-Guide-v4.4.pdf	4.4	29/08/2023
3	Validation and Verification Manual https://verra.org/wp-content/uploads/2018/03/VCS_Validation_Verification_Manual_v3.2.pdf	3.2	19/10/2016
4	VCS Project Webpage, VCS ID – 4416 https://registry.verra.org/app/projectDetail/VCS/4416	-	-
5	Initial Project Description https://registry.verra.org/app/projectDetail/VCS/4416	2.5	24/04/2024
6	Validation report https://registry.verra.org/app/projectDetail/VCS/4416	0.4	25/04/2024
7	Previous Applied Methodology VMR0006 https://verra.org/methodology/vmr0006-methodology-for-installation-of-high-efficiency-firewood-cookstoves/	1.2	-
8	New methodology applied VM0050 Energy Efficiency and Fuel-Switch Measures in Cookstoves	1.0	-
9	Revised Project Description Document	1.2	20/06/2025
10	Monitoring report	1.2	30/06/2025.
11	Emission reduction calculation		Corresponding to MR
12	Onsite audit report of the VVB done end user agreements and tokens	-	17/02/2025 to 20/02/2025
13	Local Stakeholder records: Advertisement and Invitation - local authorities, National authorities NOLs, Newspaper authorities, radio Broadcast, Social media advertisement, Invitation letter to the churches, - Feed back forms - Invoices for newspaper advertisement - Detail stakeholder report - Pictures from LSC - Radio broadcast invoices - Webinar documentation	-	2022, 2023 and 2024

	- participation list grievance address form		
14	Employment records: - Employee contracts – employment contracts - SOP signed – Safe and secure workplace, Safeguards Policy, Vehicle SOP and Visitor SOP - Contractors contracts – team leaders and field assistant - Pay slips - Employee analysis sheet	-	2022, 2023 and 2024
15	Purchase order Burn Burn Supply agreement Burn new customer Registration Form 1CM	-	27/06/2024
16	Monitoring plan spreadsheet Distribution database and Power solve Distribution Ledger Stove stacking calculation sheet	-	-
17	Training: Participants training Certificate Training acknowledgement form Training Photos Training Slides Field staff training records	-	2022, 2023 and 2024
18	ICS laboratory Management System Burn email_ECO Wood Kuniokoa Testing result	-	04/03/2024
19	Baseline survey South Pacific Forestry & Environment Consultancy(SPFEC) – review on the study TEM cookstove report for highlands region Baseline survey questionnaire database	-	Same as above
20	fNRB fNRB calculation sheet SPFEC Letter to TEM – review of FnRB calculation Geo-information for natural resource and environment management – letter on fNRB	-	Same as above
21	Evidence for start date - FDYP55	-	23/09/2024
22	Government approvals LOR CCDA LOR PNGFA Baseline study approval Stakeholder consultation approval from government	-	Same as above
23	KML file for geo coordinates for project area KML file for geo coordinates for baseline area	-	-

24	No double counting declaration List of all other project active in PNG Buen agreement for no carbon rights	-	Same as above
25	Incentive mechanism document Cookoff event photographs Pot usage video	-	Same as above
26	VCS Methodology change project description Deviation	4.0	Same as above
27	VCS pipeline listing records and details	-	-
28	Remote call with PP representative for ICS distribution record maintenance software Screen shot from the data management procedure	-	Same as above
29	KPT survey results, July 2025 Photographic evidence	-	July 2025

APPENDIX 3: ABBREVIATIONS

MR	Monitoring report
PD	Project description
MP	Monitoring Period
PP	Project Proponent
PAI	Project Instances
ICS	Improved Cook Stove
VCS	Verified Carbon Standard
SDG	Sustainable Development Goals
VCU	Verified Carbon Unit
IPCC	Intergovernmental Panel on Climate Change
AQI	Acceptable Quality Level
UQL	Unacceptable Quality Level
GP	Grouped Project
CHW	Community Health Worker

APPENDIX 4: PROJECT FINDINGS

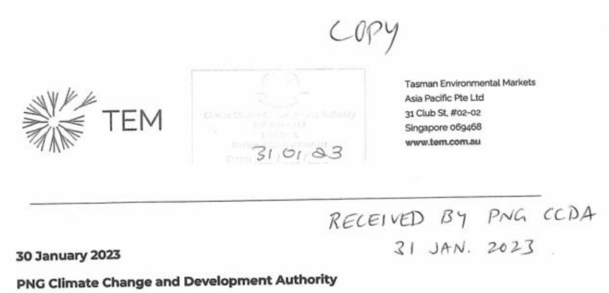
Rule	Assessment Question	Findings/Comments (CL 1)	Developer Response
VCS standard, V4.7	Additional Documents required-	1. Listing evidence and opening meeting evidence – to cross verify the chronology in section 1.4 of PD. 2. End user agreements for all the selected samples to be visited on site. 3. Distribution database covering the following details (date of sale, Geographical scale, name telephone number, address of recipient) unique ID, model and type distributed 4 Baseline survey calculation sheet 5 monitoring survey forms 6. Manufacture specification to verify the lifespan 7. KML file for stove locations in line with VCS standard	1. For listing evidence, it shows on the project's landing page (https://registry.verra.org/app/projectDetail/VCS/4416) in the Verra registry that the project was 'open for public comment from 14/07/2023 to 13/08/2023'. For Opening meeting evidence, please refer to first official email correspondence between Earthood and TEM after the validator's contract was signed. File name: VCS 4416-RE Request for Earthood's SDVista and Gold Standard Credentials .msg 2. End-user agreements have been provided under the Sharepoint folder name: End-User Information. 3. Improved Cookstoves (ICS) Distribution database has been provided. File name: VCS 4416_PNG Cookstove Distribution 02122024.xlsx 4. Data from the Baseline survey questionnaires and firewood consumption calculations have been provided in the Sharepoint Folder: Baseline Survey Calculation - VCS 4416-Baseline Survey Questionnaires_Data_Final.xlsx - VCS 4416-BCb and BCp Calculation

Rule	Assessment Question	Findings/Comments (CL 1)	Developer Response
			Sheet_Final.xlsx 5. Monitoring Surveys were conducted using a tablet where information was relayed and stored in a cloud. The full database has been provided. File name: VCS 4416_Sampling Questionnaire phase I.xlsx 6. Manufacture specification to verify lifespan has been provided under the file name: VCS 4416- Ecoa Wood G3 with Pot Skirt Product Specifications 27.07.23 2.pdf 7. KML File has been provided under the Sharepoint folder name: VCS 4416_KML files
	Rd 2	PP has submitted the requested documents, assessment team has verified them and it has been confirmed that all the submitted documents are correct. Therefore, the finding stands closed. #closed.	

Findings from revised PDD-

Rule	Assessment Question	Findings/Comments (CL 2)	Developer Response
VCS standard, V4.7	Compliance with Laws, statutes and other regulatory framework	1. Section 1.15, States that, "Additionally, TEM maintains a continuous communication with PNG's National Authorities providing frequent updates on the project activities. As an example, on the 31/01/2023,	• CCDA's evidence of receipt is provided in: Folder name: Government Approvals and Acknowledgement File name: VCS 4416_LOR CCDA.pdf

Rule	Assessment Question	Findings/Comments (CL 2)	Developer Response
		TEM had consulted with PNGFA and Climate Change and Development Authority (CCDA). During the consultations, the draft PDD and Local Stakeholder Consultation Plan were provided to the leadership of the government bodies for their comments. However, no comments have been received" To ensure compliance with national laws and the regulatory framework in Papua New Guinea, the PP is required to provide documentary evidence substantiating these consultations. Evidence that the draft PDD and Local Stakeholder Consultation Plan were shared with the relevant authorities. Compliance with applicable laws and regulatory requirements governing project approvals and stakeholder engagement in PNG.	• PNGFA's evidence of receipt is provided in: Folder name: Government Approvals and Acknowledgement File name: VCS 4416_LOR PNGFA.pdf • There are no explicit laws around cookstove projects within Papua New Guinea. TEM has been in consultation with key national, provincial, district and local stakeholders to gain access and approval into the project area. Folder name: Government Approvals and Acknowledgement > Local Stakeholder Consultation Approval: - NOL EHP.pdf - NOL SHP.pdf - NOL WHP.pdf - NOL Jiwaka.pdf Folder name: Government Approvals and Acknowledgement > Baseline Study Approval: - Approval Pilot study SHP.pdf - Approval Pilot study WHP.pdf - Government letter_Cookstoves Project_SHP (1)-signed.pdf - Government letter_Cookstoves Project_WHP (1)-signed.pdf Folder Name: District Development Authority Support Letter > Letter Sent to Authorities: - 20240207_Imbonggu DA.pdf - 20240207_Kagua Erave.pdf - 20240207_Lalibu Pangia DA.pdf - 20240207_Mendi Munhui.pdf - 20240207_Nipa Kitubu.pdf - 20240207_Police letter 1.pdf

Rule	Assessment Question	Findings/Comments (CL 2)	Developer Response
			Folder Name: District Development Authority Support Letter > Authorities Acknowledgement of Letters - Ialibu Letter to CEO.pdf - Imbonggu DA.pdf - Kangua Erave DA.pdf - Mendi Munhui DA.pdf - Nipa Kutubu.pdf - Police SHP.pdf
	Rd 2	The PP provided supporting documentation, including files VCS 4416_LOR CCDA.pdf and VCS 4416_LOR PNGFA.pdf, along with additional stakeholder approvals. During cross-checking, it was identified that two documents lack David Tow's signature. PP shall confirm whether these documents remain valid or provide an updated version containing the necessary signatures.	The wet signatures of Mr. David Tow are present on the original copies, which are in the possession of CCDA and PNGFA. The letters submitted as evidence are acknowledgement copies of the letters submitted to CCDA and PNGFA. It is noted on the letters that this is a 'copy' as seen in the example below.  COPY TEM Tasman Environmental Markets Asia Pacific Pte Ltd 31 Club St, #02-02 Singapore 069468 www.tem.com.au 31 01 23 RECEIVED BY PNG CCDA 31 JAN. 2023 30 January 2023 PNG Climate Change and Development Authority

Rule	Assessment Question	Findings/Comments (CL 2)	Developer Response
			We have attached an email from Mr. David Tow (VCS 4416-PNG Meetings 31 January 2023.eml) confirming that the original copies of the letters with his wet signature were handed over to CCDA and PNGFA. Additional Files Provided: File: Email Confirmation from David Tow with Overview of Meetings File Name: VCS 4416-PNG Meetings 31 January 2023.eml
	Rd 3	Based on the provided response, the original documents containing Mr. David Tow's wet signatures are in the possession of CCDA and PNGFA. The submitted copies are acknowledged as valid reproductions, and an email confirmation from Mr. David Tow has been provided as supporting evidence. Therefore, the finding stands closed. #closed.	

Rule	Assessment Question	Findings/Comments (CL3)	Developer Response
VCS standard , V4.7	Double counting and participation under other GHG programs	Inline with section 2.32 of VCS standard – related to double counting, "Project proponents shall not seek credit for the same GHG emission reduction or carbon dioxide removal under the VCS Program and another GHG program." PP shall submit a signed declaration on the company letter head mentioning that the ERs generated under the project and this MP will not be double counted under any other registry.	The signed declaration is provided. Refer to file titled: VCS4416_ Letter of Undertaking-TEM PNG.pdf
	Rd 2	The Project Proponent has submitted the signed declaration on the company letterhead, as required under Section 2.32 of the VCS Standard, confirming that: The Emission Reductions generated under the project and this Monitoring Period will not be double-counted under any other registry or GHG program.The declaration is provided in the file titled "VCS4416_Letter of Undertaking-TEM PNG.pdf". Upon verification of the submitted document, this finding is considered closed. #CLOSED.	

Rule	Assessment Question	Findings/Comments (CL 4)	Developer Response
VCS standard, V4.7	Free Prior and informed consent	Section 2.1.3 states that, "During the household registration phase, the end-user's consent to participate in the project is obtained. Once the willingness to participate in the project is acknowledged, information on the house geographical location and alongside a list of prerequisite questions relating to their baseline cooking behaviours are collected. The end-user is then made aware of what the project entails, how to collect their ICS, and who they may need to contact if they have any questions before they sign off on a Household Registration Form, marking their consent to join the project activity." PP shall provide documented evidence of the Household Registration Form, including signed consent forms from household selected for an on-site visit to validate the consent collection process. Detailed step-by-step documentation of how consent is obtained, recorded, and stored should also be submitted. PP shall clarify whether there is any difference between the token number and the Stove ID assigned to the Improved Cookstoves distributed under the project.	The records of household registration forms including the signed consent to participate in the project for the households selected for the on-site visit are provided as evidence. (refer to Sharepoint Folder: End-User Information) **Steps of obtaining consent:** • TEM's has multiple operational field teams and each of them consists of 4 members (1 team leader, 2 field assistants and 1 cookstove ambassador). These teams visit each household following Free, Prior Inform Consent (FPIC protocols) to seek consent from end users to participate in the project. • Once consent is obtained, the details of the end user are collected digitally by the field assistants via a tablet and recorded in a Household Registration Form and a token flyer is provided that has information on project benefits and a unique token number. • The project benefits and the contents of the End user Agreement are translated and explained by the field team during the house visit. • The End user Agreement was translated and explained a second time by TEM's field team during the cookstove handover to the end user before signing. **Token Number and Unique Serial Number** The unique token number provided to the households during the time of registration and the unique serial

Rule	Assessment Question	Findings/Comments (CL 4)	Developer Response
		If a distinction exists, the PP shall provide a detailed explanation of the allocation process, including supporting records that demonstrate how each stove is uniquely tracked within the project database.	number (USN) are different identifiers used to ensure that only one improved cookstove is distributed to a household. A Token flyer with a unique token number is provided to the end user after consent is obtained to participate in the process and the household registration step is completed. This token flyer exchanged for an improved cookstove during the distribution of the improved cookstove after the end user is verified with the photo taken during registration. The USN of the improved cookstove is logged against the token number in the data management database, which ensures that only one device is allotted per household. Please also see response to the CL below on distribution process for a more detailed explanation and evidence. This information has been updated in Section 2.1.3 of the PD.
	Rd 2	After reviewing the submitted evidence, including: Signed Household Registration Forms & Consent Documents (for households selected for the on-site visit). Training Records demonstrating FPIC compliance and field team preparedness. Verification of the Data Collection Software used to track household participation and stove distribution.	

Rule	Assessment Question	Findings/Comments (CL 4)	Developer Response
		Cross-check of Token Numbers & Unique Serial Numbers (USNs) within the data management system to ensure no duplications. The provided documentation fully substantiates the compliance of the household registration and consent collection process. With all required evidence collected and reviewed, this finding is now considered closed. #Closed.	

Rule	Assessment Question	Findings/Comments (CL 5)	Developer Response
VCS standard, V4.7	End user agreement	1. Section 1.5.1- PP has submitted an end user agreement, However, it has been observed that: The agreement is written in English, raising concerns about whether all end-users fully comprehend its terms and conditions. No supporting evidence has been provided on how end-users were made aware of the clauses and obligations stated in the agreement.	Before approaching end users, field coordinators attend a 4-day training course where they are introduced to: • The project activity and how it will benefit the end user. • How to use the ICS safely and how to maintain it. • How to conduct the Household Registration process. • The roles of the field teams and different reporting lines. • How to conduct the ICS Distribution process. These topics will be delivered using three methods: 1. Theory using PowerPoint presentations

Rule	Assessment Question	Findings/Comments (CL 5)	Developer Response
		PP shall: Provide a detailed explanation of how it was ensured that end-users understood the contents of the agreement, particularly for non-English speakers. Submit evidence (e.g., translated versions, verbal explanation protocols, training sessions, or acknowledgment records) demonstrating the measures taken to ensure end-user awareness of the agreement's clauses. Confirm whether alternative language versions or interpretation support were made available to the end-users during the agreement process.	2. Hands on activities for interviewing and explaining the project activity. 3. Simulation and Testing through group presentations The 4-day training was led by Dr. Gabriel Machovsky, Director of International Projects, and Michael Kisombo, In-Country Manager. While Gabriel conducted the training in English, Michael translated the information to tok pisin, the local language of the area. Michael also assisted to explain the training material in a more relatable form as needed. At the end of the training period, the members of the field team familiarize themselves with the content though hands on practical exercises tok pisin to simulating how they would interact with end users during the household registration and the ICS distribution phases. During both phases, tok pisin is used to communicate with end users, and only when end users understand the clauses and obligations stated in the agreements they will proceed with their signature.
	Rd 2	PP has provided an explanation regarding the measures taken to ensure end-user understanding of the agreement. However, the following aspects require further clarification and supporting evidence: Submit evidence of how the agreement was explained	During the 4-day training course (VCS 4416-PNG Staff Training 4 days.pdf): • Field assistants were provided training on the ethical considerations of obtaining informed consent (Slide 44 of training PowerPoint and

Rule	Assessment Question	Findings/Comments (CL 5)	Developer Response
		to end-users Confirm whether a written translated version of the agreement exists. If not, provide proof of verbal explanation procedures (e.g., training slides, transcripts, field reports).	VCS4416_Field Assistant Simulation Manual.pdf). • The translated contents of the Household Registration Form (VCS 4416-Household Registration Form in pidgin.pdf) and End-User Agreement Form (VCS 4416-End User Agreement Form in Tok Pidgin.pdf) were used for a hands-on training exercise (Slide 37 of training PowerPoint and VCS4416_Field Assistant Simulation Manual.pdf) where field assistants would explain the contents of the agreement with each other to practice. The training also focused on how to effectively communicate with participants, explaining the cookstoves project, and obtaining consent (Slide 46-53 of training PowerPoint). Special emphasis was placed on cultural appropriateness and clear communication, which was reinforced through: • Practical sessions to ensure staff members were confident in the consent process. • Re-training sessions to reinforce best practices and ensure consistency in the approach. • Use of Tok Pidgin and local dialects: We ensured that any potential language barriers were minimized by providing training in Tok Pidgin and any local dialects, to ensure full understanding among participants. These enabled the Field Assistants to better understand the context and importance of the FPIC and carbon transfer processes. It has also helped them better explain these concepts to end-users that are characterized by low literacy levels.

Rule	Assessment Question	Findings/Comments (CL 5)	Developer Response
			Additional Files Provided: File: Household Registration Form (Tok Pidgin) File Name: VCS 4416-Household Registration Form in Tok pidgin.pdf File: End User Agreement Form (Tok Pidgin) File Name: VCS 4416-End User Agreement Form in Tok pidgin.pdf File: Hands on activities list for Day 3 of training File Name: VCS4416_Field Assistant Simulation Manual.pdf
	Rd 3	Based on the provided response, the measures taken to ensure end-user understanding of the agreement have been adequately explained. Supporting documentation, including translated forms and training materials, has been provided as evidence of the efforts to communicate the agreement effectively. Therefore, the finding stands closed.	

Rule	Assessment Question	Findings/Comments (CL 5)	Developer Response
		#closed.	

Rule	Assessment Question	Findings/Comments (CL 6)	Developer Response
VCS standard, V4.7	Applicability of methodology	1.PP is required to revisit the applicability condition number 8 of the methodology under Section 3.2 and demonstrate its applicability. Additionally, further clarification is needed regarding the incentive mechanisms designed to reduce the use of inefficient baseline devices and practices. Demonstration of Applicability: The PP shall review and validate the applicability condition number 8 of the methodology under Section 3.2. 2. PP shall explain in detail the applicability condition, of the methodology, "The project proponent designs incentive mechanisms to reduce the use of inefficient baseline devices and practices that can be replaced by the project devices and describes these mechanisms in the project description" The PP shall submit a clear and structured action plan outlining the specific measures taken to ensure the gradual phase-out or reduction of baseline stove usage in the project activity.	1. Applicability condition number 8 and its applicability is added to Section 3.2 of the PD. 2. Project incentive mechanisms to reduce the use of inefficient baseline devices and practices are explained below: - Field assistants known as 'Cookstove Ambassadors' will actively interact with end-users from their allocated jurisdictions to encourage the use of ICSs and continuously foster awareness on the benefits of using ICS over open fire cooking methods. - Monitored results will be tabulated quarterly, and the village with the most progress adopting from open fire cooking methods will be awarded with a 'Cook-Off Event'. - The Cook-Off Event is an engagement strategy in the form of a celebration within the village where TEM will conduct activities highlighting the ICSs' benefits such as the reduction of firewood consumption and the reduction in smoke generated during cooking. - Cookstove Ambassadors will also present successful stories through knowledge sharing sessions with the event's participants. Additionally, TEM will work with Cookstove Ambassadors seeking successful testimonials

Rule	Assessment Question	Findings/Comments (CL 6)	Developer Response
		Evidence of awareness programs, financial incentives, user engagement strategies, or policy measures supporting the transition should be provided.	from end-users benefiting from clean cooking practices through the use of ICSs that can be converted into short form videos to capture broader social media audiences.
	Rd 2	PP has provided an overview of the incentive mechanisms implemented to reduce the use of inefficient baseline devices and encourage ICS adoption. However, further clarifications and supporting evidence are required to demonstrate the effectiveness and implementation of these mechanisms. While the response describes the incentive mechanisms, no documentary evidence has been provided to confirm their implementation. Additionally, PP shall clarify: How is the effectiveness of these mechanisms will be measured Are there any follow-up surveys, interviews, or tracking mechanisms to assess end-user transition from baseline devices?	As the project activity recently started, the end-users have been using their ICSs only for a short period of time. Thus, TEM is concentrating efforts on conducting awareness programs with the end-users and broader communities to strengthen their knowledge on the benefits of using ICSs. The Field Assistants' training was focused on the development of skills to interact with end-users in a professional manner while helping them to understand the benefits of cooking and maintaining their ICS correctly. These lessons have been compiled in the training manual: • Field Guide (VCS 4416- Awareness Program Field Guide.pdf) Key teachings are then summarized into Awareness Program Notes (VCS 4416_Burn Cookstoves user manual_Tok Pidgin.pdf) for Field Assistants to refer when conducting awareness programs in their villages.

Rule	Assessment Question	Findings/Comments (CL 6)	Developer Response
			As the project activities recently commenced (100 days at point of monitoring), we continue in the 'Awareness Programs' phase, to make sure that end-users are aware of the benefits gain while using ICS. Incentive mechanisms will start during the second monitoring period. Monitoring Effectiveness of Incentive Mechanism 1. The effectiveness of the mechanism will be relied on the allocation of individual wards to field assistants for them to easily monitor familiar areas, enabling them to collect data and report on cookstove's stacking. Data will be subsequently curated, and the overall staking factor will be estimated. 2. The effectiveness of the incentive scheme can be determined through a decrease in the cookstove stacking factor from the first monitoring period. Additional Files Provided: File: Field Guide File Name: VCS 4416- Awareness Program Field Guide.pdf File: Field Notes

Rule	Assessment Question	Findings/Comments (CL 6)	Developer Response
			File Name: VCS 4416_Burn Cookstoves user manual_Tok Pidgin.pdf
		PP has provided the information as well as sufficient evidences to demonstrate the effectiveness of the program therefore, the finding stands closed. #closed.	

Rule	Assessment Question	Findings/Comments (CL 7)	Developer Response
VCS standard , V4.7	Applicability of methodology	In line with methodology requirement, Project proponents implement a method for the distribution and identification of project devices that avoids double counting of emission reductions by other mitigation actions and includes unique product identification on the stove itself at the time of distribution/sale (e.g., program logo, alpha/numeric ID, and end-user location, such as geographic coordinates, complete address information)" However, the current documentation does not provide sufficient details on the procedures for unique identification and how USNs are recorded and tracked to avoid data entry errors. The PP shall provide a detailed description of the	The steps followed during distribution of the Improved cookstoves, and records maintained to ensure that only a single cookstove is distributed per household are: **STEP A: Household registration** 1. TEM's has multiple operational field teams and each of them consists of 4 members (1 team leader, 2 field assistants and 1 cookstove ambassador). These teams visit each household following Free, Prior Inform Consent (FPIC protocols) to seek consent from end users to participate in the project. 2. Once consent is obtained, the details of the end user are collected digitally by the field assistants via a tablet. Information about the project benefits and the contents of the End user Agreement (as a requirement to join the project to be signed during the cookstove distribution) are translated and explained to the end user in detail. The Household Registration Form" is signed by the end user interviewed, and a

Rule	Assessment Question	Findings/Comments (CL 7)	Developer Response
		process used to assign USNs to each ICS at the time of distribution. Evidence should be provided to demonstrate that each stove has a unique identifier that prevents duplication or multiple claims for the same device. The PP shall explain the procedure used to log USNs into the electronic database. Measures taken to prevent human errors during data entry or stove transfer shall be described, including any validation checks or automation processes.	‘Token flyer’ showing a unique ‘Token Number’ is provided in exchange. 3. As people in the project area lack a National ID, a photo of the end user signing the “Household Registration Form” is taken along with the ‘Token flyer’ such that the unique Token Number is visible in the photo. 4. The end user is instructed to be present in person with the Token flyer at the distribution location at a specified date and time for the handover of the Improved Cookstove. 5. All the steps above are captured digitally and catalogued in a data management program and converted into pdf for each individual household. **STEP B: Cookstove distribution** 6. The Improved cookstoves are shipped by the manufacturer (Burn, Kenya) with a Unique Serial Number (USN) that is attached to the cookstove body. 7. At the distribution location, an end user who has successfully signed the household registration form and is in possession of a valid ‘Token flyer’ approaches TEM’s field team to collect their allocated improved cookstove. 8. A field assistant verifies that the end user with the ‘Token flyer’ is the same as the person in the photo taken during house registration. 9. In the event the original end-user is unable to collect the stove on the distribution day, said end-user may delegate another family member to collect the cookstove on their behalf. At the point of collection where the faces of the collector does not match the recipient’s face, the matter is escalated to the team leaders for their approval to release the cookstove.

Rule	Assessment Question	Findings/Comments (CL 7)	Developer Response
			10. The field assistant enters the USN of the improved cookstove provided to the end user against the ‘Token Number’ in the data management program. 11. The field assistant thoroughly re-explains the “End User agreement” to the household member as it was originally introduced during house registration during the explanation of project benefits and requirements to join the project. 12. Each “End User agreement” includes the details of the household, the USN of the improved cookstove received, the photo of the end user with the token flyer and the company name. 13. This ensures that a single Improved Cookstove is assigned to a single Token Number and a unique “End User Agreement”. As the Token Numbers are issued in sequence, and the photo verification is done at the time of distribution, there is no duplication or multiple claims for the same device. Screenshots from the data management system, examples of photos taken for validation during registration are provided as evidence. Folder Name: End-Use Information> VCS 4416-Batch 1 Day 1 Folder Name: End-Use Information> VCS 4416-Batch 1 Day 2
	Rd 2	PP has provided a detailed explanation of the unique identification process for improved cookstove distribution and tracking. The following aspects have been sufficiently addressed:	

Rule	Assessment Question	Findings/Comments (CL 7)	Developer Response
		Unique Serial Number (USN) Assignment & Tracking:Each ICS has a USN attached at the time of manufacture. USNs are recorded against a Token Number in a digitized data management system. The End User Agreement includes USN details, household information, and a photo of the end user for verification. Measures to Prevent Double Counting & Duplication: Photo verification at the time of stove distribution. Sequential issuance of Token Numbers to prevent multiple claims. Approval system for cases where a representative collects the stove instead of the registered end user. Data Entry Validation & Error Prevention: PP has confirmed that an internal system is in place for error correction and validation during USN entry. The internal system was reviewed via a online meeting as well. The system ensures that duplicate entries are flagged and reviewed to maintain data accuracy.	

Rule	Assessment Question	Findings/Comments (CL 7)	Developer Response
		Based on this, finding stands closed. #Closed.	

Rule	Assessment Question	Findings/Comments (CL 8)	Developer Response
VCS standard, V4.7	Applicability of methodology	Inline with the recent Clarification received from VERRA related to methodology: 1. PP shall revisit the parameter BCb,i,y to update the frequency of the survey	The frequency of updating for BCb,i,y is updated to 5 years in line with the recent clarification in the PD and MR.
	Rd 2	1. PP has updated the frequency, therefore, the finding stands closed. 2. For parameter; $BC_{ex\text{-}ante,b,i}$ During the review of the Controlled Cooking Test (CCT) Logs, it was observed that the "# people consuming meal" field is marked as "NA" for some households. Since this parameter is essential for calculating per capita firewood consumption, its absence may impact the accuracy of baseline fuel consumption (BC_i) calculations. Clarify how missing household member data ("NA") was handled in BC_i calculations. Additionally, Provide justification for the 12.36% conservativeness deduction and how it was determined. Clarify the methodology used for cross-checking external data sources and how they influenced the final BC_i value.	**BCi calculations** • The baseline survey questionnaire captured the number of adults, elderly and children in each of the selected households. • From the households interviewed from the baseline questionnaires, CCTs were conducted where the equivalent household size (using a factor of 0.8 for elderly, 0.9 for adults and 0.5 for children) was available. • The CCTs were conducted in the allocated cooking areas of these households during normal meals to ensure minimal disruption. Each meal was consumed by the members of the family. • Therefore, in those data collection points where the number of people consuming the meal is missing, the equivalent household size was used instead to estimate the firewood consumption (BC_i). **Conservativeness Deduction** • The parameter that needs to be demonstrated for relative precision of 10% is the average daily firewood consumption per adult equivalent, following Section 8.2.1.1 VM0050 v1.0.

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<tr><th>Rule</th><th>Assessment Question</th><th>Findings/Comments (CL 8)</th><th>Developer Response</th></tr>
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If the relative precision is larger than 10%, then a conservativeness deduction should be applied following Section 4 of CDM Sampling and Surveys Standard v9.0.
The lower bound of the 90% confidence interval is applied, as it is more conservative for the mean daily firewood consumption per adult equivalent obtained from 71 household measurements for open fire cooking (CCTs).
Referring to '90-10 confidence' sheet of <u>VCS 4416-BCb and BCp Calculation Sheet_Final.xlsx</u>, the relative precision for the following set of inputs is of 12.36% (cell M19).

Mean – 3.3739 kg/adult eq per day
Standard deviation – 2.1081
Sample size – 71 households
Population size – 873,095
Confidence/Precision level – 90/10

This calculation is further demonstrated using the standard CDM sample size calculator (CDM-EB67-A06-GUID v03.1)
<table>
<tr><th colspan="3">Calculator to check if the precision has been met or not after a sampling survey is conducted</th></tr>
<tr><th>Input</th><th>Value</th><th>Notes</th></tr>
<tr><td>Sample size</td><td>71</td><td>Number of samples (open fire cooking)</td></tr>
<tr><td>Sample mean</td><td>3.3739</td><td>Mean of firewood consumption per adult equivalent</td></tr>
<tr><td>Sample standard deviation</td><td>2.1081</td><td>Standard devation of the measurements</td></tr>
<tr><td>Standard error of the mean</td><td>0.2502</td><td></td></tr>
<tr><td>t-value</td><td>1.6669</td><td>t-value associated with confidence level</td></tr>
<tr><td>Abosolute precision</td><td>0.4170</td><td></td></tr>
<tr><td>Relative precision</td><td>12.4%</td><td></td></tr>
</table>
(Refer to file <u>VCS4416_Precision calculation - BC_i parameter.xlsx)</u>
Referring to '90-10 confidence' sheet of <u>VCS 4416-BCb and BCp Calculation Sheet_Final.xlsx</u>
The lower bound value for average firewood consumption per adult equivalent is obtained by using the lower bound value (cell O19)

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Rule	Assessment Question	Findings/Comments (CL 8)	Developer Response
			(3.37 x (1 – 12.36%) = 2.96 kg/adult eq/day) o This is scaled by the average household size (4.85, cell Q22) to obtain the value for annual consumption (BC_i) = 5.24 tons per year (cell M35) **Cross checking with external data sources** • The value of 5.24 tons of firewood consumption per year per household was then compared with available literature and publications in section 3.10 of the baseline survey (TEM 2025_Cookstoves Highlands region report_v3_FINAL.pdf). • The value for BC_i used (5.24/household/year) falls on the lower end of the range reported in the following publications (3.64 to 11.10 tons/household/year) o The region is characterized by the lowest average temperatures in the country that leads to highest consumption of firewood in PNG (Hanson et al., 2001; ACIAR, 2013). o Murphy (2009) estimated the average firewood consumption for the Eastern Highlands Province as 3.64 tonnes/household/year. o ACIAR (2013) proposed that 7.51 tonnes /household/year was the average firewood consumption for Papua New Guinea based on data collected from the Highlands region (Eastern Highlands Province, Western Highlands Province and Chimbu Province), National Capital District and Morobe Province.

Rule	Assessment Question	Findings/Comments (CL 8)	Developer Response
			◦ Page et al., (2016) estimated an average of 11.10 tonnes/household/year for the Morobe Province • Additionally, the baseline report and pilot study were peer-reviewed by Dr. Cossey K. Yosi, an independent in-country forestry expert (Head of Department of Forestry, PNG University of Technology) who noted that the reports and measurements "followed internationally accepted protocols for such studies and are justified". Additional Files Provided: File: Precision calculation for BC_i File Name: VCS4416_Precision calculation - BC_i parameter.xlsx
	Rd3	PP has submitted the detailed calculation evidence for the calculation of parameter BCex-ante,b,I, and with the evidence for conservative calculation of the parameter. The assessment team has verified the details. therefore, the finding stands closed. #closed.	

Rule	Assessment Question	Findings/Comments (CAR 1)	Developer Response
VCS standard, V4.7	Does the project meet the template guidelines?	PP shall revise the template requirement to meet the VCS template requirement: 1. The file name shall be as per format - VCS PD ProjectID DDMMMYYYY 2. Font size of the project report shall be - size 10.5, black, regular (non-italic) Arial or Franklin Gothic Book font. 3. As per the template guidelines, "Where a section is not applicable, explain why the section is not applicable (i.e., do not delete the section from the final document" PP shall not delete any table from the project template. 4. PP Shall remove all the template guidelines from the PDD 5. Table 1.11 of the PDD mentions incorrect calendar year, PP shall revisit the section. 6. PP shall not add any additional section to the template, Under section 1.18, pp Shall revisit against the template guidelines 7. PP shall also add the guidelines related to sampling in section 3.1	1. File name will be renamed: VCS PD 4416 14FEB2025 2. Front size has been amended to size 10.5, black, regular (non-italic) Franklin Gothic Book font. 3. Additional language has been added to clearly state that the section is not applicable to the project activity. Table from the original project template has been added back into the section. 4. Template guidelines removed. 5. Table has been amended to reflect calendar year, taking into consideration the project start date of 23/09/2024. Calculation of ERs by calendar year has been included in the most updated ER Calculation Sheet for ex-ante calculations. 6. Additional sections have been removed. 7. Guidelines have been included in Section 3.1.
	Rd 2	PP has addressed all identified deviations from the VCS template requirements and implemented the necessary revisions. The following corrections have been made:	

Rule	Assessment Question	Findings/Comments (CAR 1)	Developer Response
		The file name has been revised to VCS PD 4416 14FEB2025, aligning with the required format. The font has been amended to size 10.5, black, regular (non-italic) Franklin Gothic Book, meeting the VCS template specifications. All previously removed template tables have been added back, ensuring compliance. Sections that were not applicable have been retained with appropriate justifications instead of being removed. All guideline texts have been removed from the final version of the PDD. The calendar year has been corrected in alignment with the project start date (23/09/2024).The calculation of ERs by calendar year has been updated in the ER Calculation Sheet.	

Rule	Assessment Question	Findings/Comments (CAR 1)	Developer Response
		Additional sections have been removed, ensuring compliance with the original template structure. Sampling guidelines have been appropriately included in Section 3.1. Therefore, the finding stands closed.	

Rule	Assessment Question	Findings/Comments (CAR -2)	Developer Response
VCS standard , V4.7	Dates of CP correct throughout the report?	During the review of the Project Design document, it was observed that the crediting period dates (23-September-2023 to 22-September-2034) are inconsistent with other sections of the PD document. As per the VCS Standard v4.7, project documentation must maintain consistency in crediting period dates across all sections to ensure accuracy in emission reduction claims and compliance with validation and verification requirements.	The crediting period dates are amended to 23-September-2024 to 22-September-2034 throughout the document.
	Rd 2	PP has Corrected the CP dates through out the report, there fore the finding stands closed. #Closed.	

Findings from MR -

Rule	Assessment Question	Findings/Comments (CL 9)	Developer Response
VCS standard, V4.7	Project location	In line with the VCS standard V4.7, section 3.11.1, 3) For projects with multiple project activity instances (see Sections 3.6.4 –3.6.22) and grouped projects (see Section 3.6.10), either: a) A geodetic coordinate for each instance, provided in a KML file; or b) Geodetic polygon(s) provided in a KML file that: i) Encompass all instances in the project, and ii) Delineate the smallest administrative division of land for the local government (e.g., if the activity takes place within six villages, the six villages must each have their own polygon) PP shall provide a KML file containing either: Geodetic coordinates for each project activity instance, or Geodetic polygons delineating the project boundary and covering all relevant administrative divisions.	The KML files for the project area, with provinces, districts and local level governments (LLGs) which are the smallest administrative division in PNG are provided that encompasses all instances of the project. Folder Name: VCS 4416_KML files
	Rd 2	The Project Proponent (PP) has provided the required KML files containing: Geodetic polygons covering all project activity instances, including provinces, districts, and local-level governments (LLGs), which represent the smallest administrative divisions in Papua New Guinea (PNG).	

Rule	Assessment Question	Findings/Comments (CL 9)	Developer Response
		The submitted KML files align with the requirements outlined in VCS Standard v4.7, Section 3.11.1 (3) for projects with multiple activity instances. With the submission of the complete and verifiable geolocation data, this finding is now considered closed. #Closed.	

Rule	Assessment Question	Findings/Comments (CL 10)	Developer Response
VCS standard , V4.7	Sustainable development Goals	In line with the template guidelines for Cumulative contribution over project lifetime, "the cumulative contributions over the project lifetime in the "Contributions Over the Project Lifetime" column in Table 1 below. The cumulative impact should be calculated by summing the current project contributions with all impacts included in previously approved VCS monitoring reports or Sustainable Development Contribution Reports." PP shall revisit the table and substantiate the table inline line to the requirements mentioned. SDG 1.1 - Goal 1: Amount of money saved by families that buy firewood – The PP shall quantify and explain how the project's	The following explanation line has been included: Changing from traditional open fire cooking practices to a more efficient improved cookstove will result in the reduction of firewood consumption that is usually purchased at high prices, hence reducing the economic burden of families that buy firewood under the project activity.

Rule	Assessment Question	Findings/Comments (CL 10)	Developer Response
		activities contribute to reducing household expenditures on firewood. Provide a clear methodology and data sources used to estimate the amount of money saved by families due to the adoption of the project's improved cookstoves.	
	Rd 2	The Project Proponent has correctly incorporated the cumulative contribution over the project lifetime in alignment with the template guidelines and VCS monitoring requirements. However, regarding SDG 1.1 – Goal 1 (Amount of money saved by families that buy firewood), the PP has stated that: "Changing from traditional open fire cooking practices to a more efficient improved cookstove will result in the reduction of firewood consumption that is usually purchased at high prices, hence reducing the economic burden of families that buy firewood under the project activity." While this provides a qualitative explanation, quantitative justification is still required. PP shall provide a calculative justification for the same.	The parameter to be monitored for Goal 1 in the validated PD is number of cookstoves distributed. This is because when changing from traditional open fire cooking practices to a more efficient improved cookstove, it will result in the reduction of firewood consumption that is usually purchased at high prices. Hence this reduces the economic burden of families that buy firewood under the project activity. To supplement this, a calculation was done using the baseline survey report (File name: VCS 4416-TEM 2025_Cookstoves Highlands region report_v3_FINAL.pdf), which allowed us to estimate the average savings per household from the average money spent.

Rule	Assessment Question	Findings/Comments (CL 10)	Developer Response
			It was estimated an average of 540 Kina per year per household is spent on firewood. Based on these calculations, it is estimated that 1.04 Kina is saved per household per day, and from proportioning the number of households that purchase their firewood from the monitoring survey, an estimated **57,890 kina** is saved during this monitoring period across all households. A full in-detailed calculation is presented in the sheet 'SDG Target 1.1' in the VCS 4416-SDVISTa -Monitoring Plan Spreadsheet v1.2.xlsx.
	Rd3	PP has provided sufficient evidences to demonstrate the contribution towards the SDG1, therefore, the finding stands closed. #closed.	

Rule	Assessment Question	Findings/Comments (CL 11)	Developer Response
VCS standard, V4.7	Stakeholder Identification	Legal or customary tenure/access rights PP shall confirm that the project does not harm or disobey any rights related to territories and resources	TEM ensures safe and appropriate working conditions for all its employees through the development and consultation of its Project Risk Assessment (Refer to file name: VCS 4416_PNG Cookstove Project Risk

Rule	Assessment Question	Findings/Comments (CL 11)	Developer Response
		associated with indigenous communities. Working condition section of MR mentions – PP mentions, “Support of local police, private security teams and the involvement of community leaders in ICS distribution planning enables the mitigation of identified security risk for on-ground teams” The PP shall provide detailed information on how security risks are assessed and managed during project activities. Describe the specific protocols in place to protect on-ground teams, particularly focusing on measures taken for women employees involved in the project. Verification of Safe Working Conditions: PP shall explain how appropriate working conditions are ensured for all employees, including women. Submit documentation (e.g., security policies, risk assessments, training protocols, or community engagement plans) demonstrating adherence to safety standards.	Assessment_090924.xlsx). Within this assessment sheet, risks related to transportation and security are identified for stakeholders including TEM staff, the consequences of risks to occur and their likelihood of it happening is stated via an impact score. Then, mitigation actions for these risks are included to bring the residual risk down to acceptable levels. Additionally, all personnel who travels into the project boundary must adhere to the tailored security travel plans that are developed with clear emergency protocols to guarantee their safety. (Refer to file name: VCS 4416_ICS distribution travel plan-week 10_Approved.pdf) The travel plan includes details for Remote Check-Ins with a designated Check-In person, full travel itinerary, emergency contact details, etc. The travel plan must be pre-approved by TEM’s Director of International Projects before they are circulated to all parties participating in the trip. All documents can be found in the Sharepoint Folder: Safety and Travel Plans

Rule	Assessment Question	Findings/Comments (CL 11)	Developer Response
			Please note that as personal information of the field team alongside emergency contact information is shown in the travel plan, this document must be kept strictly confidential.
	Rd 2	The Project Proponent has provided the required documentation and clarifications to address the finding related to: Legal or Customary Tenure/Access Rights: The PP has confirmed that the project does not harm or disobey any rights related to territories and resources associated with indigenous communities. The Project Risk Assessment document comprehensively outlines identified risks and their mitigation measures. Security Risk Assessment & Management: The PP has submitted a Remote Field Safety Plan and ICS distribution travel plan (VCS 4416_ICS distribution travel plan-week 10_Approved.pdf), which includes: Remote Check-In Procedures for field teams. Emergency response protocols with clear escalation steps. Security provisions and mitigation measures tailored for project staff, including women employees.	

Rule	Assessment Question	Findings/Comments (CL 11)	Developer Response
		The travel plan has been reviewed and approved by TEM's Director of International Projects, ensuring proper implementation. Verification of Safe Working Conditions: The PP has submitted documentation (VCS 4416_PNG Cookstove Project Risk Assessment_090924.xlsx) outlining: Risk assessment procedures identifying safety concerns for all employees. Mitigation measures to ensure a secure working environment. Compliance with safety protocols for transportation, emergency procedures, and security measures. The documents confirm that security risks are actively assessed and managed, particularly for women employees involved in fieldwork. With all required evidence submitted, reviewed, and verified, this finding is now considered closed. #closed.	

Rule	Assessment Question	Findings/Comments (CL 12)	Developer Response
VCS standard, V4.7	Repair and maintenance	The PP shall provide a comprehensive explanation of the repair and maintenance steps involved for the distributed stoves. Specify whether regular maintenance checks are required and who is responsible for conducting them (e.g., project team, technicians, or end-users). The PP shall clarify whether repair costs are covered under the project or if end-users are responsible for any associated expenses. If costs are applicable, provide details on the cost structure, repair service providers, and financial support mechanisms (if any).	The project proponent regularly visits end users throughout the project's duration to gather feedback and assess the condition of the cookstoves. These visits offer opportunities to address any questions or concerns about the project, promote the continued use of the cookstoves, and provide training on their maintenance. For more information, please refer to the TEM Grievance Redress Mechanism v1.0, for the full grievance and feedback procedure. However, there are no plans currently to replace any damaged device and they would be conservatively removed from the GHG emissions calculations. This information has been added to Section 3.1 of the MR. The project device has a useful lifespan of 10 years, and no major maintenance or repairs are anticipated during the operating period. The ex-ante ER calculations have assumed a 5% loss of operational cookstoves each year to be conservative in estimating the ERs (see file titled: VCS 4416_VM0050_Emission Reductions Calculation Spreadsheet_Ex Ante_v1.1, sheet name: Immediate Calculations, for more clarity). File Name: VCS 4416-TEM 2024_Grievance Redress Mechanism_v1.pdf

Rule	Assessment Question	Findings/Comments (CL 12)	Developer Response

	Rd 2	The Project Proponent has provided the required clarifications and supporting documentation addressing the repair and maintenance process for the distributed Improved Cookstoves .
		
		Key Verifications & Supporting Evidence:
		1. Regular Maintenance & User Support
		• The PP conducts periodic visits to end-users throughout the project to:
		  o Gather feedback on the condition of the cookstoves.
		  o Address any user concerns and provide training on proper maintenance.
		  o Encourage continued use and provide support through the Grievance Redress Mechanism (TEM Grievance Redress Mechanism v1.0).
		
		2. Repair & Replacement Policy
		• The PP confirms that there are no planned replacements for damaged devices.
		• Any non-operational cookstoves are conservatively removed from GHG emissions calculations.
		• This policy has been incorporated into Section 3.1 of the MR to ensure transparency.
		
		3. Cookstove Lifespan & Emission Reduction Assumptions
		• The project assumes a 10-year useful lifespan for each cookstove, as verified through manufacture specification.	

Rule	Assessment Question	Findings/Comments (CL 12)	Developer Response
		• To maintain conservativeness in ER calculations, a 5% annual loss of operational cookstoves is assumed. Conclusion: With all required clarifications, supporting documentation, and policy disclosures submitted and verified, this finding is now considered closed. #Closed.	

Rule	Assessment Question	Findings/Comments (CL13)	Developer Response
VCS standard, V4.7	Sampling	MR mentions, “24 households out of the 120 selected for surveying could not be visited during the field sampling dates of 16-17 January 2025, due to the members of the household not available at the time of the visit by the field team, or exceptional cases related to death of kin. Where a survey may not be completed, or where there is a non-response, the reasons are clearly documented.” The PP shall explain in detail how the survey maintained the required relative precision despite the 20% non-response rate (24 out of 120). Provide information on any adjustments made to sample size, statistical margins of error considered,	According to VM0050 v1.0, the monitoring survey should achieve 90/10 confidence precision for the proportion of devices in operation. Using the standard CDM sample size calculator (CDM-EB67-A06-GUID v03.1), the predicted sample size for the two batches in operation were calculated separately with expected proportion of devices in operation as 0.9: • Batch 1: 30 • Batch 2: 31 With an adequate oversampling of ~100%, 60 households from each batch were selected to be part of the monitoring survey.

Rule	Assessment Question	Findings/Comments (CL13)	Developer Response
		and alternative data collection strategies used to compensate for missing responses. Submit supporting documentation or calculations demonstrating the survey conclusion process and compliance with statistical requirements.	A total of 96 households were sampled (45 from Batch 1 and 51 from Batch 2) and the results of the survey for the parameter proportion of devices in operation in both batches were 1.0. This result was cross checked using the CDM Sample Size calculator to demonstrate that the relative precision achieved is less than 10% (achieved precision is 0%, no error). The CDM sample size calculation sheets for batch 1 and batch 2 are provided as evidence. (Sharepoint Folder: Monitoring Sampling) File name: VCS 4416-Sample Size Calculator_Batch1.xlsx File name: VCS 4416-Sample Size Calculator_Batch2.xlsx Therefore, as the selected number of 120 households (60 from each batch) included an oversampling of ~100% than the minimum sample size required, the 20% non-response rate did not have an impact on the required relative precision.
	Rd 2	The Project Proponent has confirmed that: The survey was designed with 90/10 confidence precision, as required under VM0050 v1.0. The CDM sample size calculator (CDM-EB67-A06-GUID v03.1) was used to determine the minimum	**Alternative Data Collection for Non-Response Cases**

Rule	Assessment Question	Findings/Comments (CL13)	Developer Response
		required sample size, and oversampling (~100%) was applied to ensure statistical reliability. 96 households were successfully surveyed, and the proportion of devices in operation was recorded as 1.0, achieving 0% error in the CDM sample size calculations. While the relative precision requirement has been met, the handling of non-response households remains unclear. Clarification Required: Alternative Data Collection for Non-Response Cases The PP shall clarify whether any follow-up attempts (e.g., additional field visits, phone surveys, or alternative data collection) were made to reduce the 20% non-response rate. If no follow-ups were conducted, the PP shall justify why additional data collection was not pursued and confirm whether this aligns with best practices. Impact of Non-Response on Other Survey Indicators While stove operation rates were successfully assessed, the PP shall confirm whether the missing 24 households impacted other monitored parameters (e.g., fuel savings, user satisfaction, or behaviour change metrics).	The geography of the project area (and the Highlands region in particular) is quite remote and logistically difficult to navigate. Most of the households are also situated in remote areas without available transportation or communication infrastructure (e.g., roads, mobile towers internet) preventing the monitoring team communicate with the end-users to ensure their availability for the survey. As this problem was anticipated and a high oversampling rate of ~100% (120 samples instead of 60 required) was employed to compensate for any non-responses, avoiding the need for a follow-up survey as it is not cost effective and logistically feasible. Alternate methods to obtain a response like phone calls were also not feasible due to lack of communication infrastructure (mobile phone coverage or internet availability). **Comparison to best practices in oversampling** Our approach, follows scientific principles and is also aligned with the CDM Standard for Sampling and Surveys v9.0 also recommends over sampling for precisely this reason – “not only to compensate for any attrition, outliers or non-response associated with the sample, but also to prevent a situation at the analysis stage where the required reliability is not achieved, and

Rule	Assessment Question	Findings/Comments (CL13)	Developer Response
			additional sampling efforts would be required. This would then be expensive, time-consuming and inconvenient" (page 7 footnote 11). The CDM Guideline for Sampling and Surveys v4.0 provides a best practice example case study where oversampling by 20% is advised (assuming and 80% response rate) in Appendix 1 (page 29). The oversampling rate of 100% employed in the project is well higher than the above example, and the relative precision is achieved. **Impact of Non-Response on Other Survey Indicators** The other indicator that is affected by the survey is the stacking estimation of 88% project stove usage (i.e. 12% usage rate of baseline stove). This parameter also achieves a relative precision of 6.2% as seen below (VCS4416_Precision calculation - stove stacking.xlsx)

Input	Value	Notes
Expected proportion, p	0.88	enter on a decimal scale
Confidence level	90%	e.g. for 90% enter 90
z multiplier	1.645	determined by confidence level
Relative precision	10%	e.g. for 10% enter 10
Population size, N	19,206	Total households in monitoring period
Predicted sample size, n	37	rounded up to nearest integer

Calculator to check if the precision has been met or not after a sampling survey is conducted

Input	Value	Notes
Actual sample size	96	Number of households sampled
Sample proportion	0.8800	Measured proportion 88% stove usage
Standard error of the proportion	0.0331	
Precision associated with a proportion	0.0544	
Relative precision	6.2%	

Rule	Assessment Question	Findings/Comments (CL13)	Developer Response
			Additional Files Provided: File: Precision calculation for stove usage File Name: VCS4416_Precision calculation - stove stacking.xlsx
	Rd3	PP has provided the explanation regarding oversampling and how the relative precision was met, It has been observed that despite non able to cover the entire sample range PP has still successfully able to meet the relative confidence and precision. All the relative evidences have been submitted by the PP. Therefore, the finding stands closed. #closed.	

Rule	Assessment Question	Findings/Comments (CL 14)	Developer Response
VCS standard, V4.7	Efficiency of project stove	In reference to Clarification Number 4 against the methodology, the PP must provide evidence ensuring that the efficiency of the stove meets the manufacturer's specifications and is free from bias or inconsistencies. The PP shall substantiate how the efficiency of the distributed stoves has been validated against the	The manufacturer's specification is supported by the results of a Water Boiling test conducted in an independent ISO/IEC 17025:2017 accredited laboratory. The test report is provided as evidence (Refer to file: VCS 4416-2024-04-03_ISO 19867 Kuniokoa KWB3.0.pdf)

Rule	Assessment Question	Findings/Comments (CL 14)	Developer Response
		manufacturer's specifications. Clarify the process followed to ensure efficiency testing is free from bias and aligns with industry standards.	According to VM0050 v1.0, the frequency of updating the efficiency of distributed improved cookstoves is 1 year. As the improved cookstoves in this monitoring period have only been in use for 3 months, the validation of efficiency of distributed stoves will be carried out at the end of the first year of operation (i.e. after September 2025) through Standard Water Boiling Test campaigns.
	Rd 2	The Project Proponent has provided the required clarifications and supporting documentation addressing concerns regarding the efficiency validation of distributed stoves against the manufacturer's specifications. Key Verifications & Supporting Evidence: 1. Independent Laboratory Testing & Compliance with Manufacturer Specifications The stove efficiency validation is supported by an independent Water Boiling Test (WBT) conducted in an ISO/IEC 17025:2017 accredited laboratory. The results confirm that the efficiency of the stoves meets the manufacturer's specifications and that the testing process was free from bias.2. Alignment with 0050 v1.0 Methodology Requirements	

Rule	Assessment Question	Findings/Comments (CL 14)	Developer Response
		According to VM0050 v1.0, the frequency of updating the efficiency of distributed Improved Cookstoves (ICS) is 1 year. Since the improved cookstoves have only been in use for 3 months, efficiency validation is scheduled for September 2025, as per methodology requirements. The PP has committed to conducting future Water Boiling Test (WBT) campaigns to ensure continued compliance. With all required clarifications, laboratory testing evidence, and compliance commitments submitted and verified, this finding is now considered closed. #closed.	

Rule	Assessment Question	Findings/Comments (CAR 3)	Developer Response
VCS standard, V4.7	Does the project meet the template guidelines?	PP shall revise the template requirement to meet the VCS template requirement: 1. The file name shall be as per format - VCS PD ProjectID DDMMMYYYY 2. Font size of the project report shall be - size 10.5, black, regular (non-italic) Arial or Franklin Gothic Book font.	1. File name will be renamed: VCS PD 4416 23SEP2025 – 31DEC2025 2. Front size has been amended to size 10.5, black, regular (non-italic) Franklin Gothic Book font. 3. CP Dates have been updated. 4. Language has been reworded, and missing tables have been reinstated.

Rule	Assessment Question	Findings/Comments (CAR 3)	Developer Response
		3. In section 1.2, PP shall mention the correct CP dates. 4. In section 1.5 - Where a section is not applicable, explain why the section is not applicable (i.e., do not delete the section from the final document and do not only write "not applicable"). Inline with template guidelines, PP shall not delete any table from the Monitoring report template	
	Rd 2	PP has addressed all the above points finding stands closed. #Closed.	

Rule	Assessment Question	Findings/Comments (CAR)	Developer Response
VCS standard, V4.7	Excel Sheet	Monitoring period dates mentioned in excel sheet are incorrect – PP shall revisit the sheet.	The monitoring period dates have been amended in the excel sheet. (File name: VCS 4416_VM0050_Emission Reductions Calculation Spreadsheet_MR1_v1.1.xlsx)
	Rd 2	PP has addressed it and finding stands closed. #Closed.	

Rule	Assessment Question	Findings/Comments (CL 15)	Developer Response
VCS standard V 4.7	Project stove ID and monitoring database	During the site visit, it was observed that for one of the end users, Petric Yama, the stove ID did not match the one recorded in the monitoring database. This discrepancy raises concerns regarding the accuracy and reliability of the monitoring database. The Project Proponent shall: Clarify the discrepancy observed between the actual stove ID and the database entry. Verify the accuracy of the entire monitoring database to ensure all stove IDs are correctly recorded.	Clarification: During the site visit, Petric Yama was observed with the improved cookstove (ICS) Unique Serial Number (USN): ADYY3C. According to our monitoring database, the USN: ADYY3C belongs to Joe Patrick, who was not selected as one of the sampled households. Upon further investigation into the matter, we found that Petric Yama and Joe Patrick are related as father and son living in separate but neighbouring households. "Petric" is a common pidgin variation of the English name "Patrick". The women of the households regularly cook together and had their cookstoves accidentally mingled during a communal cooking event. Evidence for Confirmation (photos taken 27 Feb 2025): 1. **Confirmation that Stove USN: ADYUNE belongs to Petric Yama** File Name: Petric Yama holding ADYUNE cookstove.jpg. This household was visited and Stove ID

Rule	Assessment Question	Findings/Comments (CL 15)	Developer Response
			verified during the monitoring survey. 2. **Confirmation that Stove USN: ADYY3C belongs to Joe Patrick** File Name: Joe Patrick holding ADYY3C cookstove.jpg 3. **Picture of Petric Yama (father) & Joe Patrick (son) with their respective cookstoves** File Name: Petric Yama (father) & Joe Patrick (son) holding cookstoves.jpg Additional Notes and References: 1. **End-User Information for Petric Yama** Folder Name: End-User Information > Petric Yama File Names: - VCS4416-118261_Photo_BeneficiaryProduct_Petric Yama.jpg - VCS4416-Device_EndUserAgreement_Png_ADYUNE.pdf - VCS4416-Device_HouseholdRegistration_Png_ADYUNE.pdf The abovementioned photographs submitted as evidence are aligned with the ones taken by our field team during the household registration and the cookstove distribution 2. **End-User Information for Joe Patrick** Folder Name: End-User Information > Joe

Rule	Assessment Question	Findings/Comments (CL 15)	Developer Response
			Patrick File Names: - 118262_Photo_BeneficiaryProduct-Joe Patrick.jpg - Device_EndUserAgreement_Png_ADYY3C.pdf - Device_HouseholdRegistration_Png_ADYY3C.pdf This can be compared with the most recent confirmation photos of Joe Patrick (above) with his cookstove.
	Rd 2	The assessment team further investigated this matter, and upon investigation it was observed that geocordinates for both Patric YAMA and Joe Patric are nearly same. Upon confirmation of that an online interview of both Joe pattric and Pattrick yama was conducted to verify there relationship as Father and son and there stove ID were also verified, it was verified both the families often cook together therefore, there stoves got exchanged. PP also submitted appropriate evidence related to the monitoring survey conducted earlier with time stamped photos to verify the details and later it was confirmed that both the stoves are functional, therefore, the finding stands closed. #closed.	

Rule	Assessment Question	Findings/Comments (CL 16)	Developer Response
VCS standard V 4.7	Operational rate of the project stove	During the site visit, it was observed that in one of the households, the end user keeps switching between the baseline stove and the project stove. This raises concerns regarding the sustained adoption and exclusive use of the project stove, which is critical for ensuring the expected emission reductions. Additionally, the 100% stove usage rate reported in the monitoring database needs to be substantiated, given the observed dual usage. Action required: Substantiate the reason for the observed switching between the baseline and project stove. Demonstrate how it ensures continued and exclusive use of the project stove, as per the monitoring and reporting framework. Demonstrate the methodology used to claim a 100% stove usage rate, including data collection, verification procedures, and any supporting evidence.	Behavioural Change: It was documented in the monitoring survey that some households are still using the improved cookstove along with the baseline cooking methods. This has been quantified and adjusted in the stove usage rate estimation (please see detailed explanation below). To encourage the continued use of the ICS, Field Assistants, also known as Cookstove Ambassadors, will routinely visit end-users throughout the project lifetime to collect feedback, maintain the project cookstoves, and address questions or queries regarding the project activity to encourage the use of the project cookstoves. Additional Incentive Strategies: TEM will also incentivise end-users and stakeholders through its Cookstove Ambassadors, who will actively interact with end-users from their allocated jurisdictions. The goal will be to continuously foster awareness on the benefits of using ICS over open fire cooking methods. Monitored results will be tabulated quarterly, and the village with the most progress adopting from open fire

Rule	Assessment Question	Findings/Comments (CL 16)	Developer Response
			cooking methods will be awarded with a 'Cook-Off Event'. The Cook-Off Event is an engagement strategy in the form of a celebration within the village where TEM will conduct activities highlighting the ICSs' economic, environmental, and social benefits such as the reduction of firewood consumption and the reduction in smoke generated during cooking. <u>Methodology Used to Claim a 100% Usage Rate:</u> TEM's has claimed a 100% usage rate of the ICSs distributed in reference to whether they are daily used. During the monitoring survey, 100% of end-users reported using their ICS daily. However, TEM has also taken into consideration stove-stacking. During the monitoring survey, end-users were also asked if they still use the Open Fire Cooking (OFC) method, and a positive answer will lead to an additional query on the frequency of this pattern.

Rule	Assessment Question	Findings/Comments (CL 16)	Developer Response
			A proportion of total meals cooked using the OFC method was tabulated, and the average of all samples collected during the monitoring period was used as the stove stacking factor. During the monitoring survey, we estimated an 12% the meals cooked in conjunction with the ICSs distributed as part of the project activity and OFC. As a conservative measure the final Emission Reductions (ERs) were discounted by 12% accounting for stacking in the project scenario, leading to 1,860 tCO2e for this first monitoring period. Evidence for Easy Reference: 1. VCS Monitoring Plan Results and Analysis: File Name: VCS-Monitoring Plan Spreadsheet.xlsx
	Rd 2	The assessment team has verified the details and evidence submitted by the PP, it has been observed that based on the monitoring survey result PP has applied 100% usage rate the same result was also observed during the onsite visit conducted by the assessment team, however, considering the project is in initial stage and PP still using baseline stove on some meal a 12% has been discounted, this 12% deduction has also been calculated inline with	

Rule	Assessment Question	Findings/Comments (CL 16)	Developer Response
		monitoring survey results. Therefore, the finding stands Closed. #closed.	

APPENDIX 5: COMPETENCE OF TEAM MEMBERS AND TECHNICAL REVIEWERS

COMPETENCY STATEMENT	
Name	Indrapal Parmar
Years of Experience	16+
QUALIFYING ROLES	
Team Leader	Yes
Auditor	Yes
Country Expert	Yes, CE Region 1 (English) / CE Region 6 (Hindi)
Technical Expert	Yes (1.1, 1.2, 3.1,13.1, 13.2)
Financial Expert	Yes
Independent Reviewer	Yes
APPROVAL	
Approved by	Head of Quality and Compliance, SustainCERT
Date	15/07/2022

Competency Statement	
Name	Muskan Chawla
Years of Experience	2.5+
Qualifying Roles	
Team Leader	Yes
Auditor	Yes
Country Expert	Yes, CE Region 1 (English) / CE Region 6 (Hindi)

Technical Expert	Yes (1.2, 3.1)
Financial Expert	No
Independent Reviewer	Yes
Approval	
Approved by	Head of Quality and Compliance, SustainCERT
Date	5/9/2023

Competency Statement	
Name	Hayden Wagia
Years of Experience	-
Qualifying Roles	
Team Leader	No
Auditor	Yes
Country Expert	Yes, Papua New Guinea
Technical Expert	Yes (3.1)
Financial Expert	No
Independent Reviewer	No
Approval	
Approved by	Head of Quality and Compliance, SustainCERT
Date	03/02/2025